



DEDICATION

THIS WORK IS DEDICATED
TO TCHOUA'S FAMILY

ACKNOWLEDGEMENTS

The accomplishment of this great project won't have been a success without out the important contributions of a good number of people who gave me a huge support academically, professionally, financially and even spiritually. I then direct my appreciations to:

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LIST OF ABBREVIATIONS

2TUP: Two Track Unified Process

API: Application programming Interface

AICS: African institute of Computer Science

CD-ROM: Compact Disk- Read Only Memory

CRUD: Create Read Update Delete;

DBMS: Database Management System

ICT: Information and Communication Technology

IOT: Internet of things

SDLC: System Development Life Cycle

ABSTRACT

Humanity has stopped evolving. In fact, it has evolved at an exponential rate over the past few centuries, driven by innovations in technology. From the agricultural revolution to the industrial revolution then to information and communication technology revolution, we have developed new tools that have allowed us to transform our world in ways that were unimaginable. Thus, we cannot see the future without technology. These new technological advances have had a profound impact on our lives, changing the way we work, the way we communicate and the way we interact with the world around us. For this reason, the African Institute of Computer Sciences is committed to making Cameroon a key player in the information and communication technology revolution. To achieve this, AICS offers its students practical trainings through academic internship. This internship gives students the opportunity to apply their theoretical knowledge to real-world problems. In line with these perspectives, we approached SMART-TECH, which is an enterprise that hosts our internship. During our induction phase, we were assigned the theme “**Conception and Implementation of an Intelligent School Management System for Student Performance Analysis and Learning Orientation** “. Following our theme, we conducted feasibility studies, followed by analysis, then via conception and realization phases. The application was developed using Node.js, Vue.js and in order to obtain a flexible and secure solution.

Keywords:

- ARTIFICIAL INTELLIGENCE
- IOT
- SCHEDULING
- COMMUNICATION

RÉSUMÉ

L'humanité n'a pas cessé d'évoluer, bien au contraire, elle a évolué à un rythme exponentiel au cours des derniers siècles, stimulée par les innovations technologiques. De la révolution agricole à la révolution industrielle, puis à la révolution des technologies de l'information et de la communication, nous avons développé de nouveaux outils qui nous ont permis de transformer notre monde de manière inimaginable. Ainsi, nous ne pouvons pas concevoir l'avenir sans technologie. Ces avancées technologiques ont eu un impact profond sur nos vies, changeant la façon dont nous travaillons, communiquons et interagissons avec le monde qui nous entoure. Pour cette raison, l'Institut Africain d'Informatique (IAI) s'est engagé à faire du Cameroun un acteur clé de la révolution des technologies de l'information et de la communication. Pour atteindre cet objectif, l'IAI offre à ses étudiants des formations pratiques à travers des stages académiques. Ce stage donne aux étudiants l'opportunité d'appliquer leurs connaissances théoriques à des problèmes du monde réel. Dans cette optique, nous nous sommes adressés à SMART-TECH, une entreprise qui accueille notre stage. Lors de notre phase d'induction, nous avons été assignés au thème " **Conception et Mise en Œuvre d'un Système de Gestion Scolaire Intelligent pour l'Analyse des Performances des Élèves et l'Orientation de l'Apprentissage** ". Suivant notre thème, nous avons effectué des études de faisabilité, suivies d'une analyse, puis des phases de conception et de réalisation. L'application a été développée en utilisant Nodejs, Vue.js afin d'obtenir une solution flexible et sécurisée.

Mots-clés :

- Intelligence artificielle
- Internet des Object
- Planification
- communication

GENERAL INTRODUCTION

In today's educational landscape, institutions face growing challenges in managing academic activities efficiently while providing personalized support to students. Traditional school management systems focus mainly on administrative tasks such as scheduling, grading, and data storage, often neglecting performance analysis and individualized learning guidance. This gap limits timely interventions and hinders continuous academic improvement. To overcome these limitations, this project presents the **conception and implementation of an intelligent school management system** that leverages artificial intelligence to analyze student performance and provide personalized learning orientation. The system seeks to combine administrative efficiency with AI-driven decision support, offering real-time insights and data-based recommendations to enhance subject mastery and overall academic success. The project's analysis and design adopt the **2TUP (Two-Track Unified Process)** methodology, emphasizing both business and system modeling for a comprehensive understanding of user needs and system functionalities. Using **UML (Unified Modeling Language)** diagrams—such as use case, class, sequence, and activity diagrams—the system is modeled to represent its behavior, user interactions, and data structures. This structured analytical approach establishes a solid foundation for subsequent design and implementation phases.

Project Phases:

- ✓ **Insertion Phase:** Defines project context, objectives, and overall purpose.
- ✓ **Existing System:** Manual, slow, decentralized, and prone to information gaps.
- ✓ **Specification Book:** Lists system requirements and development guidelines.
- ✓ **Analysis Phase:** Identifies user needs and system functionalities.
- ✓ **Conception Phase:** Designs system architecture and component interactions.
- ✓ **Realization Phase:** Develops and integrates the designed system modules.
- ✓ **Functionality Testing Phase:** Verifies system performance and requirement compliance.
- ✓ **The user guide:** Explains registration, navigation, and application process clearly



CHAPTER I : INSERTION PHASE

Preamble

The insertion phase document presents the internship environment, the structure of the organization, the conditions in which the interns were received, as well as the project to be realized during the internship.

Content :

INTRODUCTION

- I. WELCOME AND INTERGRATION
- II. GENERAL PRESENTATION OF SMART-TECH
- III. HARDWARE AND SOFTWARE RESOURCES OF WELLD OF SMART-TECH
- IV. ORGANIZATION OF SMART-TECH
- V. BRIEF PRESENTATION OF PROJECT

CONCLUSION



INTRODUCTION

Social and professional integration refers to the process by which an individual is integrated into the social and economic context by adopting the norms and rules that govern it. In the following sections, we will begin by describing our reception and integration within the company SMART-TECH. Then, we will explain its overall structure as well as its organizational chart.



1. WELCOMING AND INTERGRATION

2. Welcome

Upon our arrival at SMART-TECH, the reception was very simple. We were first taken to the meeting room, introduced to each team member, and then each of us introduced ourselves with our name, class, school, reason for being there, duration of the internship, and how we were admitted.

The Director then gave a welcome speech and instructed the intern coordinator to give us a full tour of the company. Finally, based on our academic fields, we were assigned to different departments

3. Integration

At first, adjusting was quite intense due to the workload and pressure of meeting tight deadlines. However, after three days, adaptation was achieved.

4. GENERAL PRESENTATION OF SMART-TECH

1. History

SMART-TECH is a company specializing in software development, online presence management, digital marketing, tech consulting, systems maintenance, IT training and consultancy, network design and implementation, and surveillance camera installation.

It was founded in 2020 by Mr. DJATCHOU TONYA Aimé, who had a futuristic vision for digital transformation in businesses. Today, SMART-TECH supports and works with over 10 companies aiming to align with digital standards. Its headquarters is in Douala, with a branch office in Yaoundé (Shell building).

2. Geographical location

SMART-TECH is located in New-Bell, 300 meters from the former building crossroads, heading toward the Batier center in New-Bell.

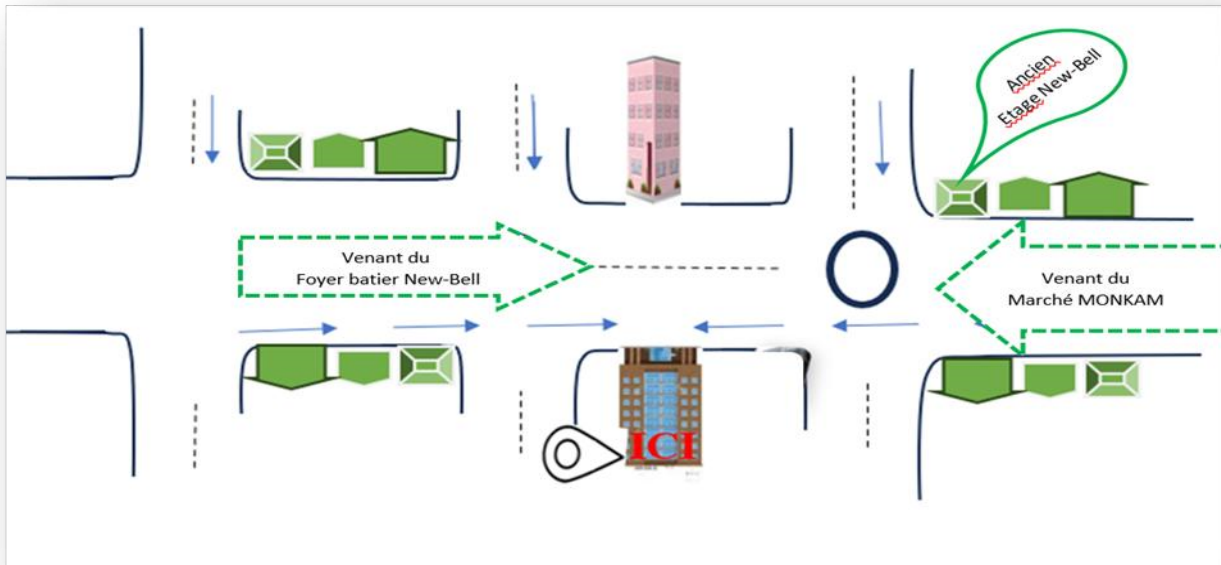


Figure 1 localisation of smart-tech

5. Mission

SMART-TECH's main mission is to promote digital transformation for individuals and businesses. Its objectives include:

- Providing a new image of Africa (especially Cameroon) to the world by helping businesses move from analog to digital to improve performance.
- Creating software, logos, flyers, websites, and network installations.
- Offering quality training to company staff and individuals, believing every African should master digital tools.

6. Activities, Achievements and Partners.

I. ACTIVITIES

SMART-TECH is a key player in digital transformation across various business sectors. Its solutions include:

- ✧ System Integration: Helping companies adapt to digital tools.
- ✧ IT Strategy and Organization Consulting: Advising on IT service organization and strategies.
- ✧ Project Management: Supporting businesses throughout project execution.
- ✧ Communication & Digital Marketing: Handling online presence, including web design and content.
- ✧ Web & Mobile Application Development: Building websites and apps with attention to quality.

II. ACHIEVEMENTS

- Design and implementation of a Travel Management System (TMS) for GHF SARL
- Network deployment and maintenance for SOCADEP, CWORDH ENERGY, and individuals
- Installation of surveillance systems at EMERIX COMPANY and private residences
- Design of logos and flyers for SMARTEKS, LA PETITE BAYAM, FONDATION PAVI, and individuals
- Online presencemanagement for LA PETITE BAYAM, SMARTEKS, PENTIUM GSM, and others
- Creation of a professional training center with JPTEKS
- Maintenance of IT equipment for FONDATION PAVI and others.

III. PARTNERS

SMART-TECH as an economic enterprise has the following partners who greatly help in the development of their enterprise.

Tableau 1 Table of partners

PARTNERS	
□ SOCADEP	
□ EMERIX.	
□ JPTEKS	
□ PENTUIM GSM	  



□ FONDATION PAVI.

IV. ORGANIZATIONAL CHART OF SMART-TECH

SMART-TECH has several departments. We were assigned to the Technology Department.

V

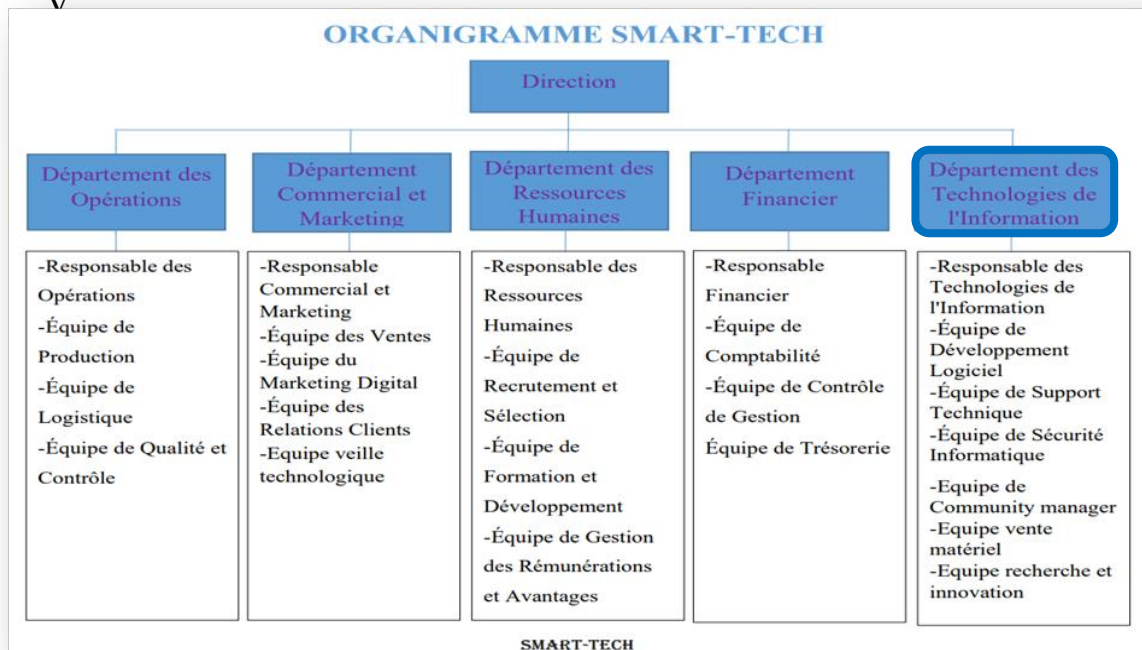


Figure 2 Organisation chart of SMART-TECH

IV. COMPANY RESOURCES

1. Hardware Ressource

Tableau 2 hardware ressource of smart-tech

N	Equipment	Characteristic	Quantity
1.	Computer sever	Desktop HP 	03
2.	Laptop computer	Laptop HP 	05
		Laptop Dell 	
3.	Routers	Cisco catalyst 1941 	01
		Tp-link (wifi) 	01
4.	Switch	Switch cisco catalyst 24 ports 	01
5.	Optic fibre	Camtel	01
6.	Modem	Modems MTN 4G 01	01
		Modems Toplink 5G	01
7.	Ethernet cable	Twisted pair CAT6	01

2. Software Resources

Tableau 3 software resources of smart-tech

N°	Software	Description
01	 Office suite	Standard Office Suite for creating documents, spreadsheets, presentations, and other professional content.
02	 Windows 11	Operating System for computer.
03	 WPS	Word Processing software.
04	 Adobe suit	Professional Suite for image editing, illustration creation, video editing, and audio editing.
05	 Canva	Online Graphic Design Platform for creating professional visuals, presentations, social media publications, etc.

06	 Visual studio code	IDE (Integrated Development Environment) , a free and open-source environment, very popular for web and mobile development.
07	 Wamp	Local Web Development Environments , including Apache, MySQL, and PHP, for testing and deploying web applications locally before putting them online.
08	 Cisco ISE	Authentication Management Tools (such as Active Directory) for managing authentications within information systems
09	 ENSP of HUAWEI	Email Monitoring Software (such as Microsoft Exchange Monitor), a useful tool for monitoring and optimizing mail server operations
10	 SolarWinds	Email Monitoring Software (such as Microsoft Exchange Monitor), a useful tool for monitoring and optimizing mail server operations
11	 EVE NG	GNS3 (or other network simulation software), a software for simulating networks through clientless virtualization.
12	 Eclipse	Open-source Integrated Development Environment (IDE) Used for software

		development in various languages, including:Java, C++, Python, PHP
	 GitHub	Source Code Management Platform , used to store, share, and collaborate on software projects.



CONCLUSION

Upon completing our integration, we have gained a thorough understanding of the work environment in which we will be operating during our internship. We have acquainted ourselves with the structure's history, missions, responsibilities, organization, various resources, and organizational chart, enabling us to confidently proceed to the second part of the report, the specifications document.



CHAPTER II: EXISTING SYSTEM

Preamble:

The insertion phase helped us to establish a comprehensive understanding of the host organization. This phase emphasizes on the key characteristics, specificities, and objectives of the designated study subject

Content :

INTRODUCTION

- I. PRESENTATION OF THEME
- II. DESCRIPTION OF THE EXISTING SYSTEM
- III. CRITICISM OF THE EXISTING SYSTEM
- IV. LIMITATIONS OF EXISTING SYSTEM
- V. PROBLEMATIC
- VI. PROPOSED SOLLUTION

CONCLUSION



INTRODUCTION

Engineers are society's problem-solvers, working to make life easier for everyone. Before proposing solutions, we need to understand the root causes of the problem and how it's currently addressed. Analysing existing systems in our field is key to identifying problems and developing effective solutions. This ensures our proposals improve upon or replace the current system for the better

PRESENTATION OF THEME

Educational institutions, though committed to providing quality education, face increasing challenges in effectively managing student performance and providing timely academic guidance. Traditional manual and semi-digital processes make it difficult to analyse student marks comprehensively, identify learning gaps, and propose orientation strategies that can help students improve in their studies.

To address this, implementing an **intelligent school management system** powered by **artificial intelligence (AI)** is essential. This system will not only streamline school operations and reduce administrative burdens but also enhance student learning by providing personalized performance analysis and orientation. It will facilitate easy access to academic data, identify strengths and weaknesses in student performance, and recommend learning strategies tailored to each student's needs.

By leveraging modern technology, schools can improve the quality of education, increase transparency in academic processes, and provide a supportive learning environment that empowers both students and teachers. This innovative solution is designed to contribute to academic excellence by combining efficient school management with AI-driven learning support.

THEME:

Conception and Implementation of an Intelligent School Management System for Student Performance Analysis and Learning Orientation

DEFINITION OF TERMS

- **Artificial Intelligence (AI):** The capability of a machine or system to simulate human intelligence by analyzing data, recognizing patterns, and making recommendations or predictions.
- **School Management System:** A software platform designed to manage and automate school operations, including administrative, academic, and communication processes.

- **Performance Analysis:** The systematic study of student marks and academic records to identify strengths, weaknesses, trends, and areas that need improvement.
- **Learning Orientation:** A personalized academic support process that guides students with tailored strategies and recommendations to improve subject mastery and overall performance.

DESCRIPTION OF THE EXISISTING SYSTEM

The current system for managing school operations, student performance, and academic orientation at ESSTIC Cameroon is still manual and fragmented, creating several inefficiencies and challenges. The existing system is characterized by:

- **Manual performance analysis:** Teachers rely on spreadsheets or paper-based methods to calculate and analyse student marks, which is slow, error-prone, and lacks depth.
- **Entrance exam management issues:** Entrance exams are handled manually, making registration, grading, and result publication tedious and time-consuming.
- **No unified platform for marks across all levels:** Academic results are stored separately at different levels (primary, secondary, high school), with no centralized database to track student progress over time.
- **No intelligent guidance:** Students do not benefit from personalized academic orientation; feedback is often generic and not adapted to individual strengths and weaknesses.
- **Limited staff management:** There is no integrated system for managing teacher records, workload distribution, and performance monitoring, leading to inefficiencies in human resource utilization.
- **Lack of centralized platform:** Information related to exams, marks, staff, and performance is scattered, making it difficult for administrators, teachers, and parents to access relevant data.
- **Limited accessibility:** Academic and administrative information is not easily accessible on multiple devices (web, mobile, desktop).

- **No real-time insights:** Changes in schedules, marks, or performance trends are not communicated instantly, delaying decision-making and interventions.
- **Absence of automated recommendations:** There is no system capable of suggesting personalized learning strategies or improvements for students based on their academic data.
- **Difficulty in tracking progress:** Long-term tracking of student performance, from entrance exams to higher levels, is inconsistent and inefficient.
- **Time-consuming processes:** Teachers and staff spend considerable time on administrative tasks (marks entry, exam preparation, record management), reducing focus on academic improvement.
- **Lack of analytics and visualization:** The absence of dashboards and analytical tools prevents schools from identifying trends, predicting performance, and making data-driven decisions.

These limitations highlight the urgent need for an **intelligent school management system** that integrates **entrance exam management, marks across all levels, staff management, and AI-driven student performance analysis with learning orientation**. Such a platform will centralize school operations, empower stakeholders with real-time insights, and provide personalized academic guidance to improve both institutional efficiency and student success.

CRITICISM OF THE EXISTING SYSTEM

The current school management practices at ESSTIC present several critical shortcomings that hinder efficiency, accuracy, and the ability to support students effectively. Some of the main criticisms include:

- **Inefficient data management:** Student marks, entrance exam results, and staff information are stored manually, leading to errors, inconsistencies, and difficulties in retrieval.
- **No centralized academic tracking:** There is no unified system to manage marks across different levels of education, making it difficult to follow a student's progression from entrance exams through higher classes.
- **Inability to provide personalized guidance:** The current system does not support AI-driven analysis or tailored recommendations to help students improve in weak subjects.
- **Limited accessibility:** Without a centralized digital platform, parents, teachers, and administrators face difficulties accessing vital information in real time and across devices.
- **Poor communication:** Communication between teachers, students, and parents remains informal (e.g., phone calls, WhatsApp), often resulting in misunderstandings and delays.
- **Inefficient resource utilization:** Teachers and staff spend excessive time on manual tasks such as mark entry, timetable preparation, and report generation, reducing productivity.
- **Security concerns:** Sensitive academic and administrative information stored physically or in isolated files is vulnerable to loss, theft, or unauthorized access.
- **Difficulty in tracking performance:** Without analytics, it is hard to monitor student progress, identify weaknesses, or make data-driven decisions for academic improvement.
- **Lack of automation:** Processes such as sending reminders, notifications, or performance alerts are manual, which slows down communication and reduces efficiency.

- **Insufficient analytics and visualization:** The current system cannot generate insights, trends, or predictions about student performance, limiting its usefulness for decision-making.
- **Limited scalability:** As student populations grow and school operations expand, the current system cannot effectively handle the increased demand.
- **Fragmented staff management:** There is no centralized system for managing teacher workload, attendance, or performance evaluation.

These shortcomings underscore the urgent need for a **modern, intelligent, and AI-driven school management system** that integrates **entrance exam management, marks tracking across all levels, staff management, performance analysis, and personalized learning orientation**. Such a system will empower ESSTIC to improve efficiency, strengthen communication, and foster academic excellence.

LIMITATIONS OF THE EXISTING SYSTEM AND PROPOSED SOLUTIONS

To overcome the criticisms of the existing system, AICS Cameroon can implement the following solutions:

Tableau 4 table of limitations of the existing system

LIMITATION	CONSEQUENCE	SOLUTION
Manual data management	Errors, inconsistencies, and difficulty in accessing student/staff/marks information	Implement a centralized digital platform for academic and administrative data management
No centralized academic tracking	Difficulty monitoring student progress across entrance exams and multiple levels	Integrate a unified system to manage marks across all levels and generate academic history
Inability to provide personalized guidance	Students receive generic feedback, with no targeted learning strategies	Integrate AI to analyze marks and recommend personalized learning orientations
Inefficient resource utilization	Time and resources wasted on manual tasks; reduced staff productivity	Automate grading, attendance, reporting, and staff workload management

Limited accessibility	Parents, teachers, and students struggle to access information in real time	Develop cross-platform applications (web, mobile, desktop) for easy access
Poor communication	Misunderstandings and delays due to informal channels	Implement a communication portal with real-time messaging and notifications
Security concerns	Risk of loss, theft, or unauthorized access to sensitive information	Secure cloud storage with encryption, authentication, and role-based access
Difficulty in tracking performance	No real-time insights into student strengths, weaknesses, or trends	Deploy AI-powered dashboards to track performance and visualize progress
Lack of automation	Inefficiencies in reminders, notifications, and reporting	Implement automated systems for alerts, performance feedback, and deadlines
Insufficient analytics	Difficulty predicting outcomes or identifying performance gaps	Develop analytics dashboards with AI to generate insights and predictive analysis
Limited scalability	Inability to handle growing student population and staff needs	Build cloud-based, scalable infrastructure to support expansion
Fragmented staff management	Lack of centralized records and workload tracking	Implement staff management modules for attendance, assignments, and evaluation
Difficulty tracking events	Missed deadlines and uncoordinated school activities	Add a centralized school calendar for exams, meetings, and events



PROBLEMATIC

How can ESSTIC improve academic management and orientation by implementing an **intelligent school management system** that integrates entrance exam management, marks tracking across all levels, staff management, and AI-driven performance analysis to provide personalized learning guidance for students?

PROPOSED SOLUTION

To address ESSTIC's challenges in managing academic and administrative processes, this project proposes the development of an **Intelligent School Management System** that centralizes data management, automates administrative tasks, and provides AI-driven insights for student performance analysis and personalized learning orientation. The system will integrate academic records, attendance, grading, and staff management, enabling real-time monitoring of student progress and identification of strengths, weaknesses, and trends. It will also include automated alerts, a communication portal, and analytics dashboards to support data-driven decisions, improve productivity, enhance communication, and provide targeted guidance for students, while ensuring secure access to sensitive information.



CONCLUSION

The study of the existing system is a very complex and challenging task. However it is an essential step in the SDLC (System Development Life Cycle). Taking the time to study an existing system permits system analyst lay a proper foundation for the new system. In this document we got to know the weaknesses, challenges, and the problematic of the existing system. Knowing these now permits us move further to implementing our new system



CHAPTER III : SPECIFICATION BOOK



Content :

INTRODUCTION

I. CONTEXT AND JUSTIFICATION OF PROJECT

II. PROJECT OBJECTIVES

III. EXPRESSION OF NEEDS

IV. PROJECT PLANNING

V. ESTIMATED COST OF PROJECT

VI. PROJECT CONSTRAINTS

VII. DELIVERABLES

CONCLUSION



INTRODUCTION

The specification book is a crucial part in software engineering process. For this reason, care must be taken when elaborating this document in order for the solutions to clearly satisfy the client's needs. This document permits us specify the expectations of the client(s), there by establishing a certain agreement between the client(s) and the person(s) to realize or develop the project. This specification book is to avoid inadequate production result



I. CONTEXT AND JUSTIFICATION

1. *CONTEXT*

The educational landscape at ESSTIC faces significant challenges in managing academic data, tracking student performance, and providing personalized learning guidance. Manual processes for record-keeping, grading, and communication result in errors, inconsistencies, and difficulty accessing student and staff information. Teachers struggle to monitor student progress across levels and exams, while students receive generic feedback without targeted learning strategies. To address these issues, this study proposes the conception and implementation of an **Intelligent School Management System** that centralizes academic and administrative data, automates routine tasks, provides AI-driven performance analytics, and supports personalized learning orientation. This system will streamline operations, enhance decision-making, and improve the overall educational experience at ESSTIC.

2. *JUSTIFICATION*

Implementing an intelligent school management system at ESSTIC is essential to improve efficiency, productivity, and educational outcomes. The proposed system will provide a centralized platform for academic data management, student performance analysis, and personalized learning recommendations. Key benefits include:

Enhanced Data Management: Improved accuracy, security, and accessibility of student and staff records.

Performance Monitoring and Guidance: AI-driven analytics provide insights into strengths, weaknesses, and tailored learning strategies.

Improved Efficiency: Automation of grading, attendance, reporting, and staff workload reduces administrative burdens.

Enhanced Communication: Secure communication portals facilitate real-time interaction among teachers, students, and parents.

Data-Driven Decision-Making: Analytics dashboards support strategic decisions to enhance academic quality and student outcomes.

II. PROJECT OBJECTIVES

i. GENERAL OBJECTIVES

The general objective of this study is to design and implement an **Intelligent School Management System for ESSTIC** that centralizes academic and administrative processes, enhances student performance analysis, and supports personalized learning orientation. Specifically, the system aims to automate data management, monitor student progress in real time, provide AI-driven insights for learning guidance, and facilitate secure communication among stakeholders, ultimately improving efficiency, transparency, and the quality of education.

ii. SPECIFIC OBJECTIVES

To achieve the general objective, the following specific objectives are targeted:

- **Centralized Digital Platform:** Develop an intuitive platform for managing academic and administrative data accessible to teachers, students, and administrators.
- **Automated Attendance and Grading:** Streamline attendance tracking and grading to reduce errors and save time.
- **Performance Analysis and Learning Guidance:** Use AI to analyze student performance, identify strengths and weaknesses, and provide personalized learning recommendations.
- **Communication and Collaboration:** Implement a secure portal for communication between teachers, students, and parents.
- **Data Security:** Ensure secure storage, role-based access, and protection of sensitive information.
- **Scalability:** Design the system to handle a growing student population and expanding academic programs.
- **Productivity Enhancement:** Reduce manual workload for staff and improve operational efficiency.

Project Characteristics:

- **Project Name:** ESSTIC Intelligent School Management System
- **Project Target:** Efficient academic data management, performance tracking, and learning orientation
- **Technical Specification:** AI-driven analytics and automated administration

III. EXPRESSION OF NEEDS

i. FUNCTIONAL NEEDS

These are the requirements that a system must meet in order to fulfill its purpose, typically expressed in terms of systems' output, input and behaviors. They are as follow:

- a. Super Admin
 - Update application
- b. Admin
 - Publish Announcement
 - Manage Timetable
 - ✓ Review timetable and validate
 - ✓ Update timetable
- c. DAAC (Department Academic Affairs Committee)
 - Assign Semester Subjects
 - ✓ Activate subjects for the semester
 - ✓ Deactivate subjects
 - ✓ View completed subjects for historical reference
 - ✓ Create new courses
 - ✓ Update course details

- Manage Availability Forms
 - ✓ Create availability forms for teachers
 - ✓ Update availability forms
 - ✓ Delete availability forms
- Update Timetable
 - ✓ Make modifications to the existing timetable
- Publish Announcement

d. Head of Department (HOD)

- View Notification List
 - ✓ Access a list of all notifications relevant to the department
- Manage Notifications
 - ✓ Create new notifications for the department
 - ✓ Update existing notifications
 - ✓ Delete notifications when no longer relevant
 - ✓ View and respond to department messages
- Edit Department Timetable
 - ✓ Make changes to the department-specific timetable

IV. NON-FUNCTIONAL NEEDS

It specifies the quality attributes of a software system. They judge the software system application based on the system's requirements, performance, usability, scalability, portability and other non-functional standards that are critical to its success. Failing to meet the nonfunctional needs can result in system not fulfilling the user's needs.

i. Usability

➤ User Interface (UI) Design

- ✓ The system should have a user-friendly interface that is intuitive and easy to navigate. Enhances user satisfaction and adoption rates among administrators, teachers, and students.

ii. Performance

➤ Response Time

- ✓ The system should respond to user actions (e.g., generating timetables, loading pages) within acceptable time limits.
- ✓ Ensures efficient use of time and reduces user frustration.

iii. Scalability

- The system should handle an increasing number of users, courses, and departments without significant degradation in performance.
- Accommodates growth and seasonal fluctuations in scheduling demands.

iv. Reliability

➤ Availability

- ✓ Requirement: The system should be available and accessible to users with minimal downtime during critical operational hours.
- ✓ Ensures continuous access to the whole information.

v. Security

➤ Data Protection

- ✓ Requirement: User data, timetable information, and system configurations should be securely stored and protected against unauthorized access or data breaches.
- ✓ Safeguards sensitive information and maintains user trust.

➤ Authentication and Authorization

- ✓ Requirement: Implement robust authentication mechanisms (e.g., cross-factor authentication) and fine-grained authorization controls to ensure that users only access information relevant to their roles.
- ✓ Prevents unauthorized access and maintains data confidentiality.

vi. Maintainability

➤ Code Maintainability

V. PROJECT ESTIMATED COST

i. SOFTWARE RESOURCES

For the realization of this project, we will be needing the following software application:

Table 1 : software resources needed

Software Resource	Description	Quantity	Price (FCFA)/day	Total Price (FCFA)
Microsoft windows10pro (64 bits)	Operating System	01	57247	57247
Microsoft office 365	It is a productivity suit that includes excel, word, outlook etc. to create documents, spreadsheets, presentations and emails	01	37247	37247
Visual Paragidm	Modeling tool	01	Freemium	Freemium
Icogram	Card-graphic tool	01	Freemium	Freemium
Visual Studio	Front-end development environment	01	Freeware	Freeware
Gantt Chart	Planification tool	01		
Node js	Back End development environment	01	Freeware	Freeware



CONCEPTION AND IMPLEMENTATION OF AN INTELLIGENT SCHOOL MANAGEMENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND LEARNING ORIENTATION (CASE STUDY: ESSTIC)



Mongo dB	NoSQL database	01	Freeware	Freeware
Flutter	Front end development environment	01	Freeware	Freeware
Postman	It is an API development environment that helps to build, test and document APIs	01	Premium	Premium
Microsoft Edge	a search engine	01	Freeware	Freeware
Total Cost Of Software Resource (FCFA)				94494

ii. HARDWARE RESOURCES

This includes the material resources used to realize the project. The table below show a representation of these different material.

Table 2 : Hardware resources needed source mercurial 2024

Software resource	Quantity	Price (FCFA)/day	Total price (FCFA)
laptop 8G of RAM, SSD, corei5, processor 1.10GHz	02	591394	1182788
Modem (Camtel)	01	50000	50000
Hard Disk Drive (500Gb)	01	50000	50000
USB Key(16Gb)	01	10000	10000
Internet Connection(MTN)	/	126500	126500
Total Cost Of Software Resource (FCFA)			1419288

iii. HUMAN RESOURCES

Table 3 : Human resource needed

Human Resource	Quantity	Price (FCFA/day)	Duration	Total Price
Head of Project	01	30000	31 days	930000

Analyst	01	25000	21 days	525000
Developer	01	20000	30 days	600000
Tester	01	10000	10 days	100000
TOTAL				2155000

VI. TOTAL ESTIMATED COST FOR THE PROJECT

Table 4: Estimated cost table

SOFTWARE RESOURCE	94495 FCFA
HARDWARE RESOURCE	1419288 FCFA
HUMAN RESOURCE	2155000 FCFA
Unexpected charges (10%) (Total cost) * 10%	1200000 FCFA
TOTAL COST	4868782 FCFA



TOTAL : 4,868,782 FCFA

i. PROJECT PLANNING

Table 5 : project Planning

Phase	Objective	Outing	Duration
INSERTION	Collection of information on the enterprise	Insertion report	14 days
EXISTING SYSTEM	Description of the existing system	Existing system	6 days
SPECIFICATION	Define the project scope, goals and functional requirements	Specification book	11 days
ANALYSIS	➤ Study of the existing system ➤ Use case and textual description ➤ modelling	Analysis document	15 days
CONCEPTION	➤ outline system architecture ➤ design the database schema ➤ design the user interface	Conception document	10 days
REALIZATION	➤ set up development environment ➤ implement system and database ➤	Realization document and source code	21 days
TESTING	➤ perform unit testing ➤ test for performance	User testing	5 days
USER GUIDE	User guide of the solution	User manual and guides	5 days

PRESENTATION	Realization of the PowerPoint and presentation of the work realized	Power point	5 days
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ii. PARTICIPANTS

The table below presence the different persons who took part in the accomplishment of this project. They include:

Table 6: table of participant

Names	Functions	Role and Tasks
Mme NZALI	Lecturer at AICS-Cameroon Software Engineer	Academic Supervisor
Mrs <i>DJATCHOU AIME</i>	entrepreneur	CEO of SMART-TECH
Mr. TCHOUA NGATCHOU NANA SAMORY	third year Student at AICS-Cameroon	Project head, analyst, conception and coding of project, testing
Mr AGBOR ANDERSON	Lecturer at AICS-Cameroon Software Engineer	Teacher

A. Gantt Diagram

The Gantt project software stands as a highly effective instrument for illustrating the diverse tasks within a project, their respective dates and the precise sequence in which they are executed.

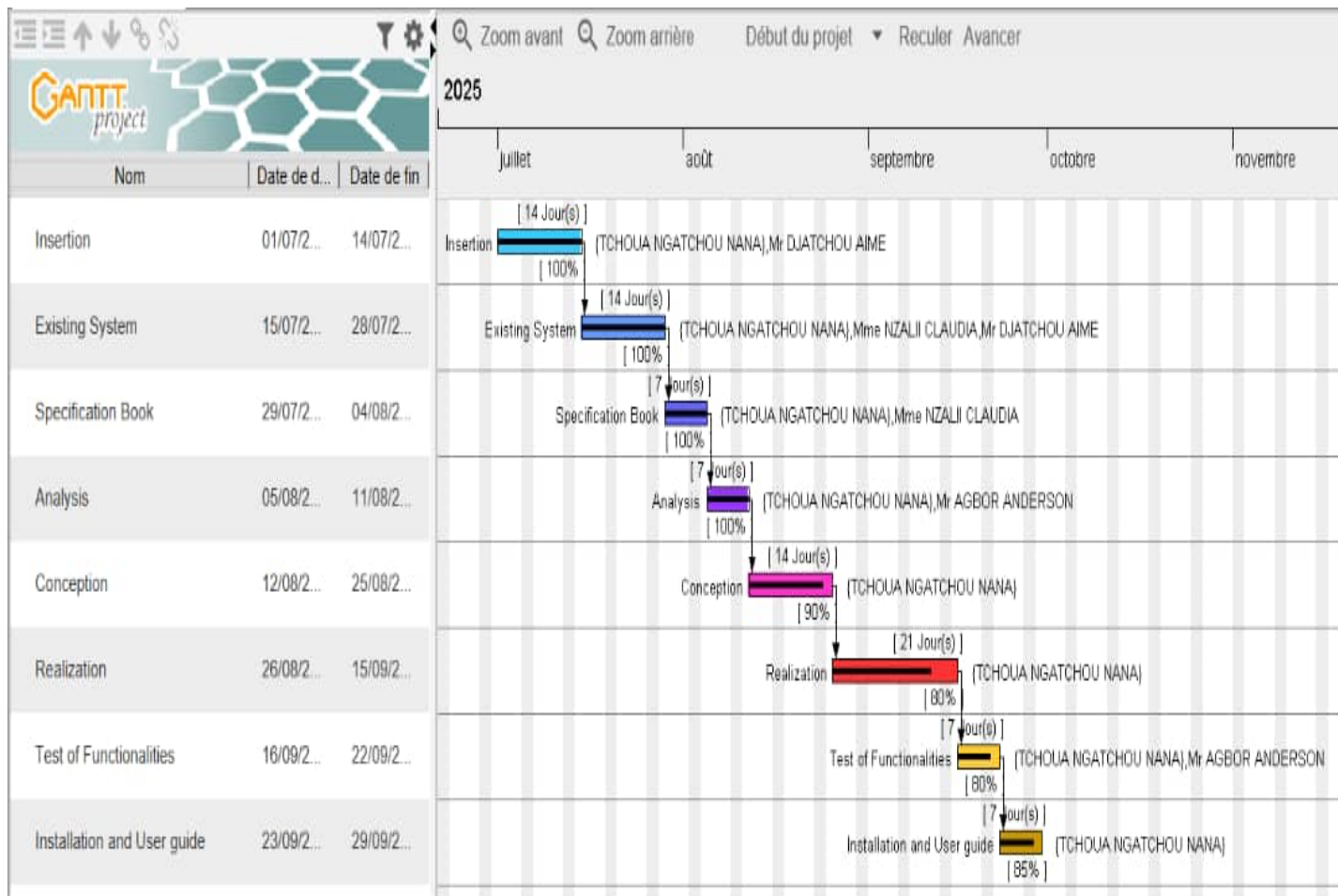


Figure 3: Gantt chart

VII. PROJECT CONSTRAINTS

To put in place this application, we must respect three main constraints which includes:

- ✓ **Budget Constraint:** the fixed budget for the realization of this project must be strictly respected.
- ✓ **The Dateline Constraint:** the realization of this system must respect the dateline attributed to it and should be completed in a duration of two months (2) including respecting the fixed objectives of the project.
- ✓ **Quality Constraint:** after fixing the budget and a dateline, we must produce a flexible, standalone and reusable application

VIII. DELIVERABLES

In this section. We will present the different elements that will be delivered at the end of this project. They include:

- A CD-ROM containing the web application;
- The installation guide and user manual of the application;
- PowerPoint Presentation of the application;



CONCLUSION

To conclude the specification book has permitted us to present the general and specific objectives of the project, the project constraints, the functional and nonfunctional need of the project, lay out the different resources of the project and finally the project planning. The end of this document will now enable us easily dive into drafting the analysis document which will bring out the technical aspects of the project as a whole. CHAPTER IV: ANALYSIS



CHAPTER IV: ANALYSIS PHASE



INTRODUCTION

Analysis is the decomposition of a problem into its various elements for computer processing. Therefore, for the successful completion of a project, it is very important to carry out a good analysis in order to achieve the objectives specified in the specification book. The realization of a good project is mainly based on its analysis. Thus, a poorly analysed project cannot give good results.



INTRODUCTION

1. METHODOLOGIE

2. Modeling

CONCLUSION

a. METHODOLOGY

Analysis is the fundamental step in the conception of a software. It is the base to the realization of every information system. An information system is an organization of resources (human, software and hardware), data and feedbacks to attain a specific objective. Several methods and languages have been put in place to facilitate the analysis and conceptions of information systems which we principally have UML and MERISE.

MERISE (Méthode d'Etude et de Realization Informatique pour les Systèmes d'Entreprise) it is a method of analysis, conception and realization of information systems mostly used in French Speaking Companies or Enterprises. It is based on the separation of data and processing them using several conceptual and physical models. Its principal objective is to conceive an information system. Merise proposes to consider the real system from two points of view I.e.

- ✓ The Static View (data);
- ✓ The Dynamic View (treatments).

It is a systematic and conceptual method of analysis.

UML (Unified Modeling Language) is a modeling language that gives the standardize visualized view of an information system. It uses diagrams to represent each aspect of a system in an object-oriented way.

UML proposes an approach different from MERISE as its readability and reusability makes it an ideal choice for programmers.

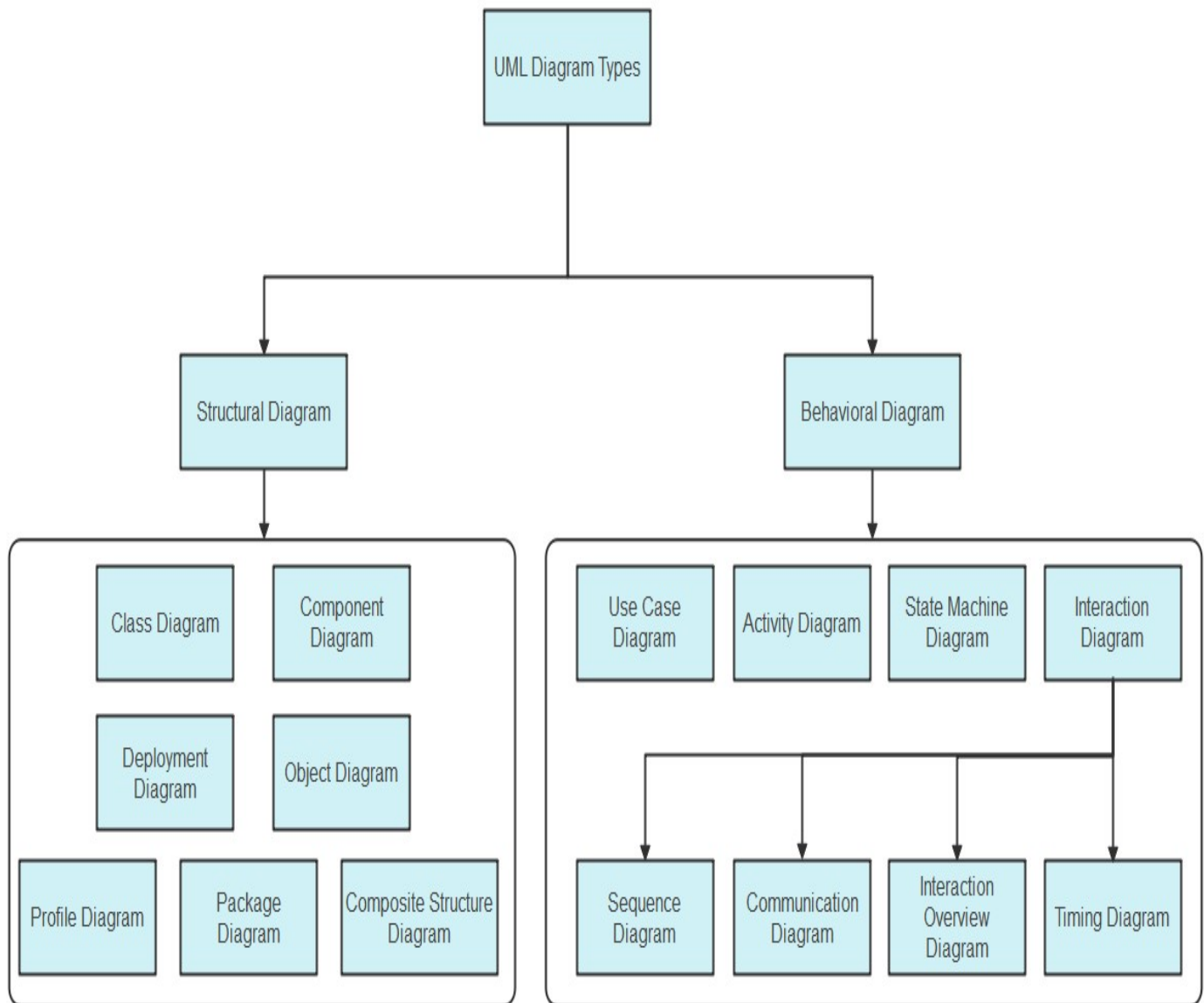
UML is currently at it 2.5 version and presents a number of advantages which includes:

- ✓ It is a formal and standardize language.
- ✓ It provides visual representation of the structure and behavior of the system which can help to communicate and understand the design of the system.
- ✓ It is flexible and customizable to suit different domain and technologies.
- ✓ It is readable and re-useable i.e. you can easily see the relationships and interactions between classes and objects in UM

- ✓ It is a planning tool that can be used to help design and document a system before the implementation phase.

Figure 4 : UML hierarchy

We can present the UML hierarchy as follows:



IX. 1.1 COMPARATIVE STUDIES BETWEEN MERISE AND UML

Table 7: Comparative study between UML and Merise

MERISE	UML
Méthode d'Etude et de Realization Informatique pour les Systèmes d'Entreprise	Unified Modeling Language
Merise is a method of analysis	UML is modeling language
Merise is a systematic method of analysis and design for information systems i.e. it uses a system approach	UML is however not a method but rather an object-oriented language, uses the 2TUP approach to model a system
It is based on procedural approach	Based on object approach
Merise proposes to consider the real system from two points of view: - The Static view(data) - The Dynamic view(treatments) Which means in Merise we have a separate study of data and treatments	UML offers a different approach from that of MERISE in that it combines data and process

A. PRESENTATION OF THE 2-TUP METHOD

2-TUP (Two Track Unified Process) is Y in shape and is a unified process which has as objective to bring a solution to constraints of functional changes imposed on information systems. It proposes a developments cycle which separates the functional aspects from the technical aspects. 2-TUP therefore distinguishes two main branches:

(The functional and technical branch) which their different results are being merged to realize our system.

Any unified process UP (Unifies Process) process meets the following characteristics:

It is iterative and incremental. Some of the advantages of the 2TUP method includes:

- ✓ It allows flexibility in the analysis of a project
- ✓ It is easy to design and understand
- ✓ It allows to have a preliminary view

Below is
the

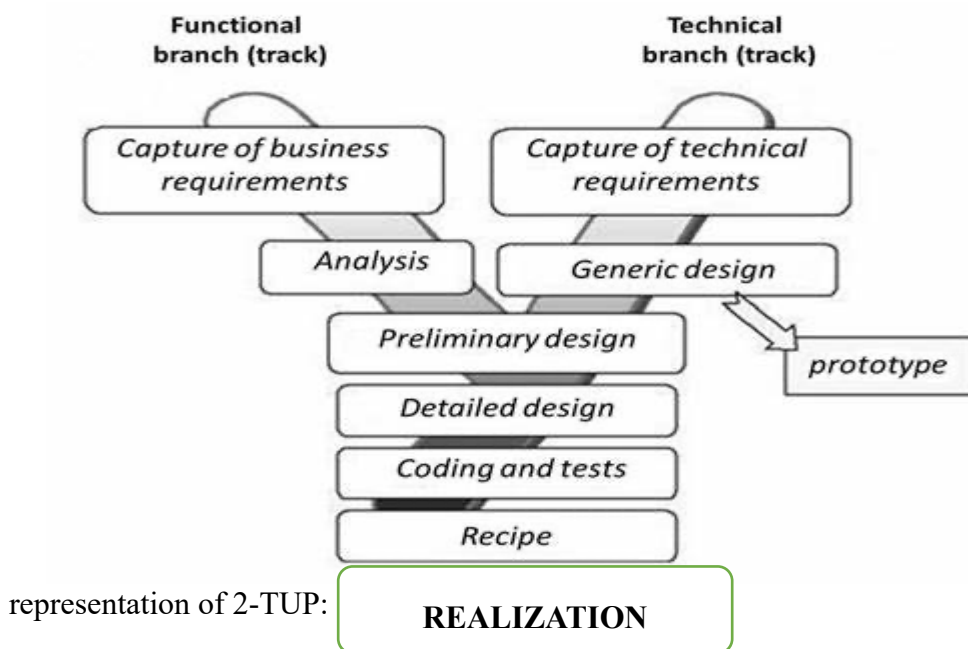


Figure 5:Representation of 2-TUP

This process is articulated with these essential phases:

- i.) **The Functional Branch:** this branch captures business needs, resulting in a model focused on the end-user's requirements.
- ii.) **The Technical Branch:** this branch proposes a generic design from the technical needs, identifies the materials, tools and technologies to be used.
- iii.) **The Implementation:** The implementation phase consists of bringing the two branches together, providing the application design and finally the delivery of a solution adapted to the needs.

a.) Functional Branch (left branch)

- ✓ **Capture of Business Needs:** this part minimizes the risk of producing an inadequate system with the needs of users and also verifies its consistency.
- ✓ **Analysis:** this is the study of the specifications to find out what the system will be made of in terms of the business rules. We have the following diagrams: Use Case, Activity Diagram, State Machine Diagram, Communication diagram

b.) Technical Branch

- ✓ **Capture of Technical Needs:** this stage consists of the identification of tool, materials and technologies to use in developing the system. This technical architecture will be presented in this stage.
- ✓ **Generic Designs:** the technical architecture will be presented in this stage. We have the following diagram: mockUp7 Design.

c.) Implementation Branch (middle branch):

- ✓ **Preliminary Branch:** this is the stage where the analysis model is integrated into the technical architecture. The goal here is to know what technical component will be used depending on the features of

the analysis. We have the following diagrams: interaction overview, Component, Deployment, package diagram and Composite Diagram.

- ✓ **Detailed Design:** this is the detailed design of each feature of the system. We have the following diagrams: class, object, sequence and timing Diagrams.
- ✓ **Coding and Testing:** This is the programming phase of the designed features alongside testing of the coded features.
- ✓ **Recipe (Result):** this is the validation phase of the functions of the system developed.

A. CHARACTERISTICS OF 2TUP

The two-track unified process was built to be used with the Unified Modelling Language and has the following characteristics:

- ✓ **It is user oriented:** 2TUP is built from user's expectation
- ✓ **It is component oriented:** it offers flexibility to the model by supporting the re-use of components.
- ✓ **It is an iterative process:** processes are being done in iterations and each iteration shows a precise level of abstraction.
- ✓ **It is an incremental process:** allowing a better functional and technical risk management and thus constituting the deadlines and the cost control.

iii. JUSTIFICATION ON THE CHOICE OF THE ANALYSIS

Our choice was based on the modelling language UML associated to it a method of analysis 2TUP due to certain criterion which are as follows:

- ✓ UML is based on object-oriented approach;
- ✓ UML associate's data and treats them;
- ✓ It is a modelling language which focus mainly on users' needs and expectations;
- ✓ 2TUP is a process built from object-oriented approach and is built from UML;

2TUP facilitates the modelling of complex systems which can further be readjusted.

2. MODELING

2.1 USE CASE DIAGRAM

i.) Definition:

The use case diagram simply shows the functionalities of the system, their interdependency and how they can be linked with actors of the system. It also clearly defines the boundaries of the system.

ii.) Formalism:

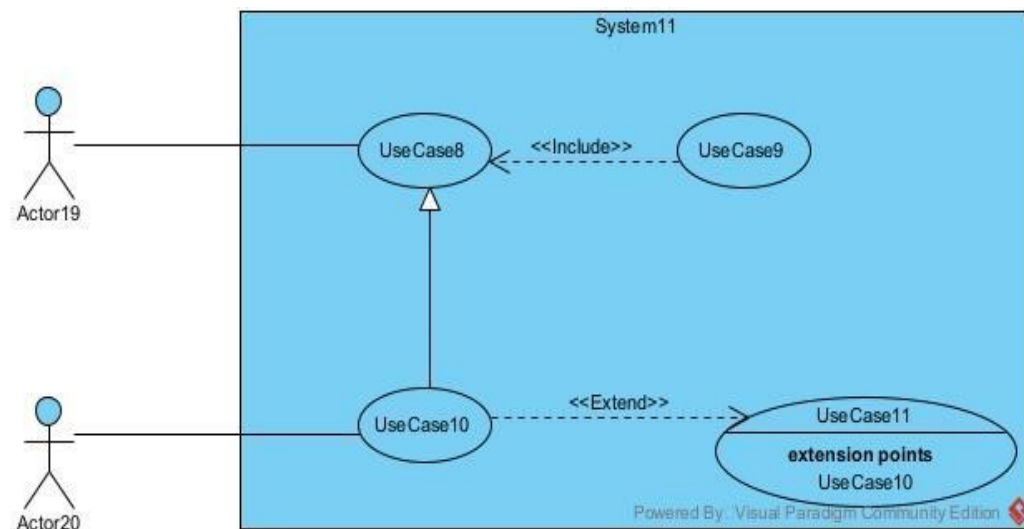
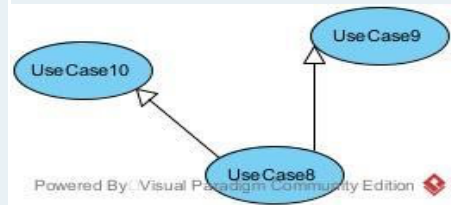
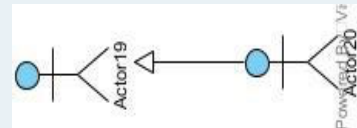
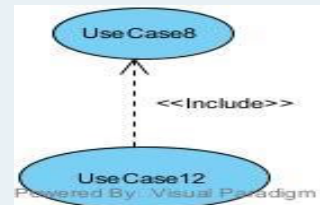


Figure 6 : Use case formalism

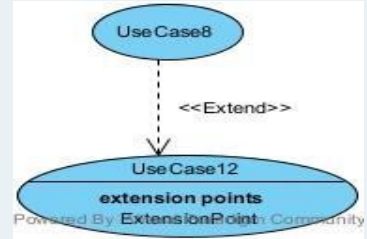
iii.) Components of a Use Case Diagram

Table 8 : Component of use case Diagram

Elements	Description and main properties	Representation
Actor	Represents the entity who directly interacts with the system. The actor is who performs the different task or actions of the system.	
Use Case	it represents a functionality of the system. It is an action that can be performed by an actor	
Generalization	It shows that a use case is a kind of another. This relationship permit the decomposition of a complex use case to simpler ones.	
Inheritance	this is the only possible relationship that can exist between actors whereby an actor can inherit from another	
Inclusion	It is denotes that an included action must be performed before another action can be done.	

Extension

It denotes that an action may be performed while another action is going on e.g. a use case B extends A implies B is an optional part of A.



iv) Graphical representation of use cases (General use case)

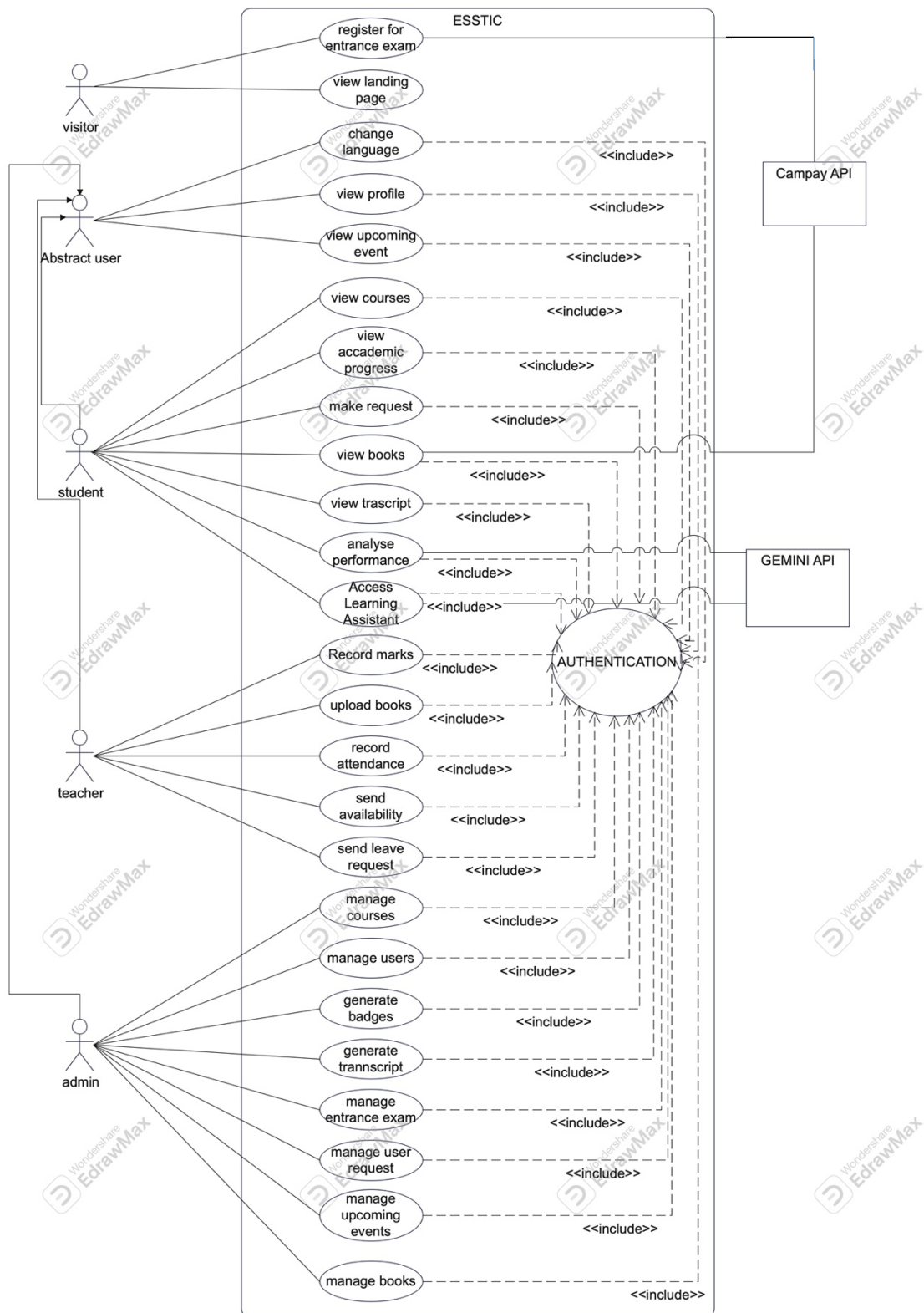


Figure 7 : General use case of the system

Manage Books (ADMIN)

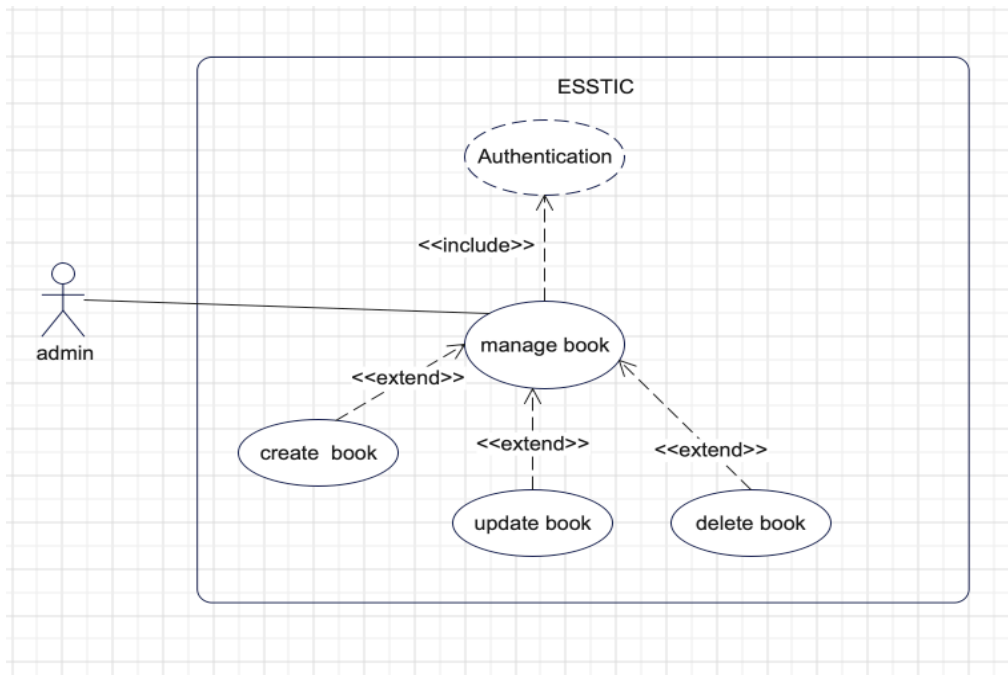


Figure 8: manage books specific use case

Manage users specific use case (ADMIN)

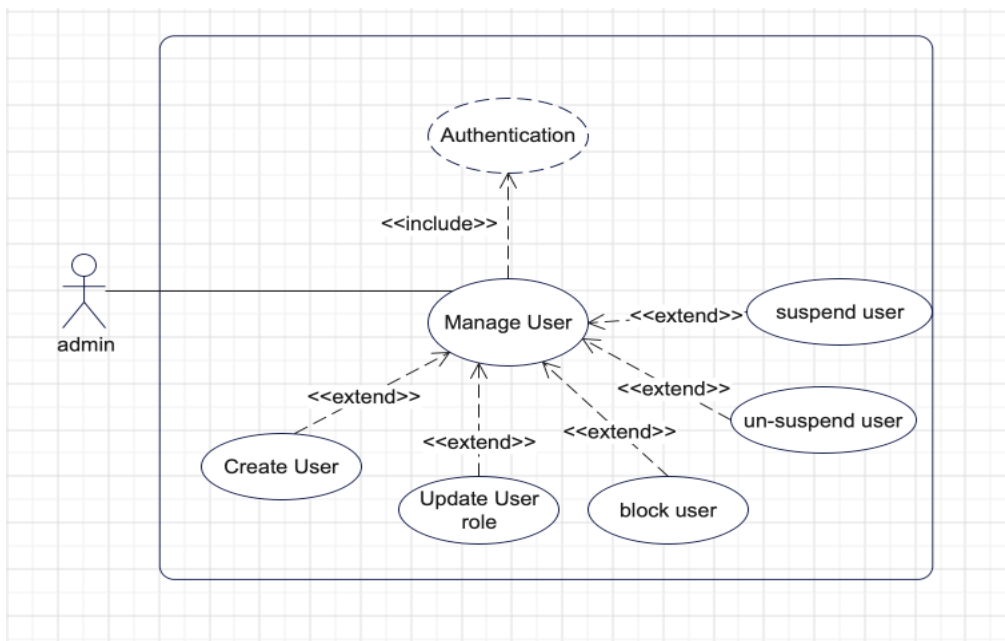


Figure 9 : Manage users specific use case

Manage Courses (ADMIN)

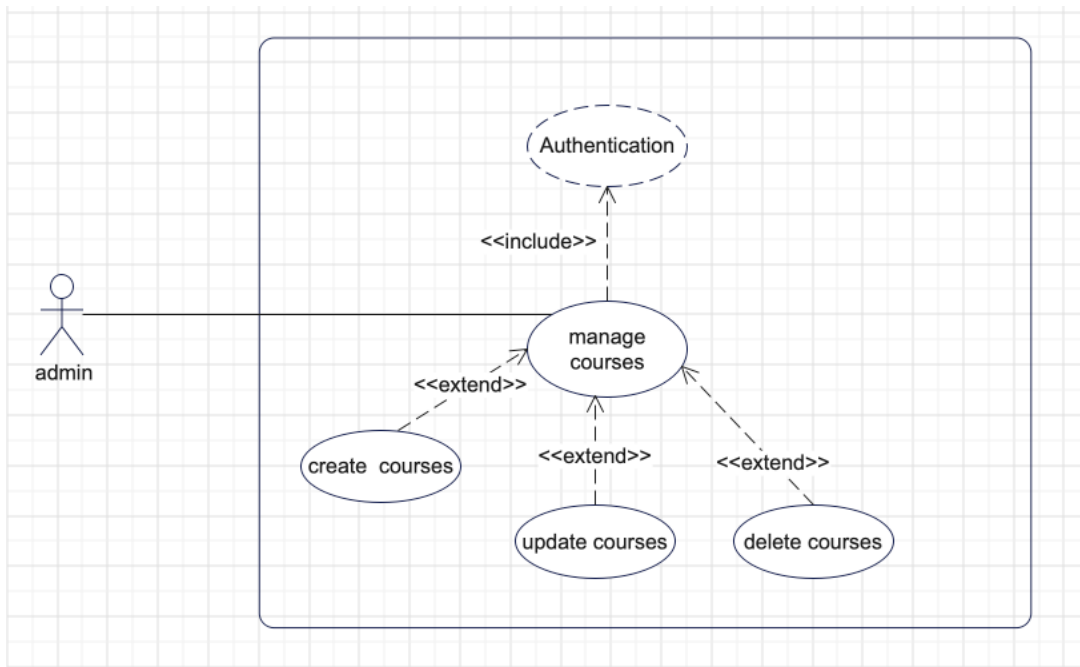


Figure 10 : manage Courses specific use case

Manage Entrance exam (ADMIN)

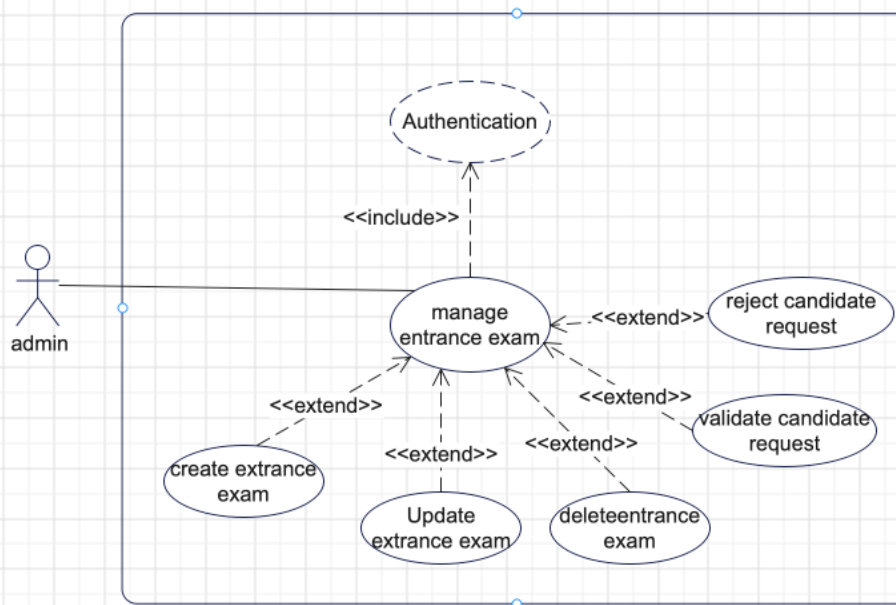


Figure 11 Manage courses specific use case

TEXTUAL DESCRIPTION OF SOME USE CASES

Every use case is associated to a series of actions representing a given function. A use case is then an abstraction of a chain of executions. Implies an instance of a use case is known as a “scenario” which is a series of steps undertaken to execute a specific use case.

The Textual description is a simple text that describes how a use case can be performed in a more precise and understandable way following a chronological order. We will present in the following lines the textual description of some use cases.

1. Use Case <Authentication>>

Table 9: Authentication textual description

Use Case « Authentication »	
Actor	Students, teachers, admins
Objective	Be able to connect to the platform and perform task
Precondition	The actor must enter the url
Trigger	1) User clicks on Authentication button.
Nominal scenario	2) System displays login form. 3) User fill and submit form. 4) Verifies form conformity 5) Send query to verify email 6) Send password to very if password match 7) system render users' page.
Alternative Scenario (invalid credentials)	8) From step (3) above if form not correct system display error message 9) From step (4) if email not found system display error message 10) From step (6) if password not matched system display error message
Exception Scenario	11) From step (4) above if there is inter connection problem system display error message
Post-condition(s)	user landing page is displayed

2. Use Case <<Create course>>

Table 10 : Textual description create course

Use Case « create course »	
Actor	Admin
Objective	Be able to create a course
Precondition	The actor enters the URL
Trigger	1) User clicks on create course button.
Nominal scenario	2) System displays create form. 3) User fill and submit form. 4) Verifies form conformity 5) Send query to verify if course exist 6) Send query to save course 7) System displays success message.
Alternative Scenario (invalid credentials)	8) From step (3) above if form not correct system display error message 9) From step (4) if course found system display error message
Exception Scenario	10) From step (5) above if there is inter connection problem system display error message 11) From step (6) above if there is inter connection problem system display error message

Post-condition(s)

Course is created

3. Use Case <<Fill Availability Form>>

Table 11 : set availability textual description

Actor	Teacher
Objective	Fill Week Availabilities.
Precondition	Actor must Authenticate.
Trigger	1) Click on fill availability form.
Nominal Scenario	2) Teacher clicks on fill availability form button. 3) System displays availability form. 4) Teacher Fill and submit form with availabilities. 5) System verifies if form was felt correctly and submit query 6) System displays success message “availabilities submitted successfully”.
Alternative Scenario (if form invalid)	7. from step (4) System display error message 8. System returns to step (3)
Alternative Scenario (if Availability already exists)	10. from step (5) System display error message 11. Return to step (3)
Post-Condition	Teacher ‘s availabilities are submitted

4. Use Case <<Create user>>

Table 12 : Generate timetable textual description

Use Case « create user »	
Actor	Admin
Objective	Be able to create a user
Precondition	The actor must authenticate
Trigger	1) User clicks on create user button.
Nominal scenario	2) System displays create form. 3) User fill and submit form. 4) Verifies form conformity 5) Send query to verify if course exist 6) Send query to save course 7) System displays success message.
Alternative Scenario (invalid credentials)	8) From step (3) above if form not correct system display error message 9) From step (4) if course found system display error message
Exception Scenario	10) From step (4) above if there is inter connection problem system display error message 11) From step (5) above if there is inter connection problem system display error message
Post-condition(s)	Course is created

X. 2.2 COMMUNICATION DIAGRAM

i. Definition

The communication diagram is used to model the specific interaction between the participants of the system. It describes both the static and dynamic behavior of the system. It shows the chronological succession of the actions performed by an actor in order to realize an operation.

ii. Formalism

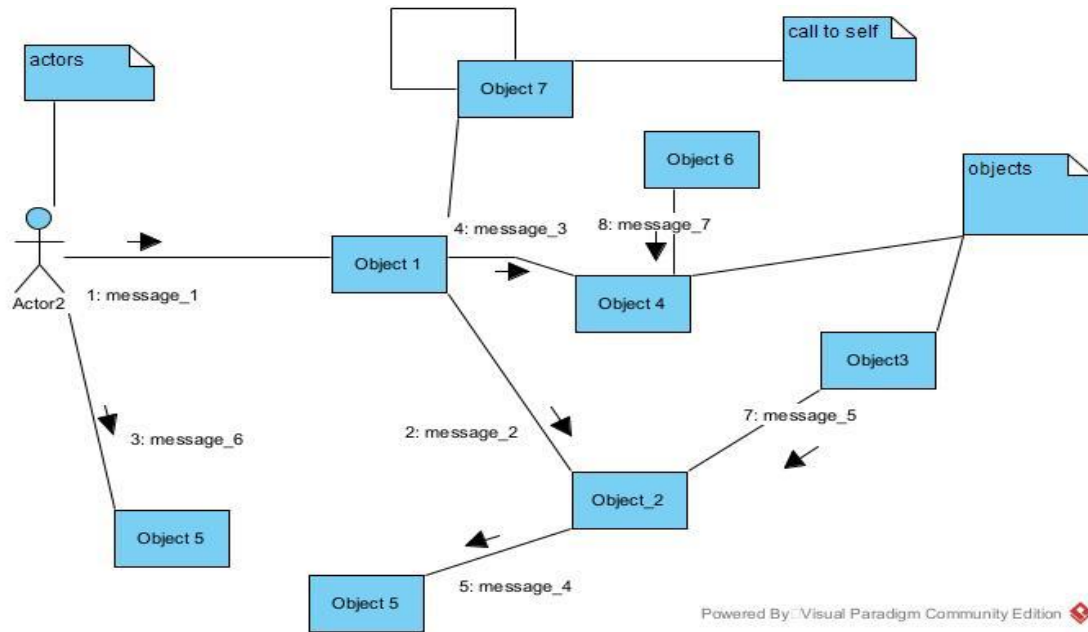
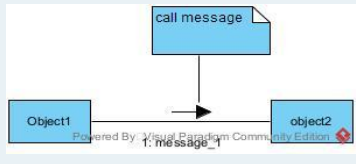
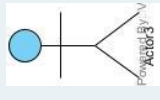
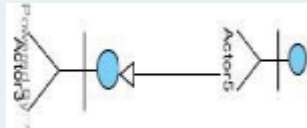
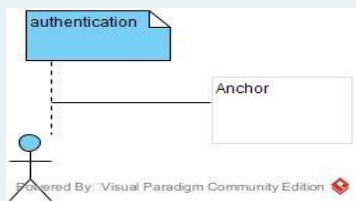


Figure 12: Communication diagramme formalism

iii. Components of a Communication Diagram

Table 13 : Component of communication diagram

Element	Description and properties	Representation
Object	Represents a role or object instance that participated in the interaction model.	

Call Message	A message that represents invocation of operation of target object	
Actor	It models the type of role played by an entity that interacts with the subject	
Generalization	Exist between two relationships where one is specialized form of the other	
Anchor	A connector that connects to a note shape for attaching additional description of connected model element	

iv. *Communication diagram for <<Authentication>>*

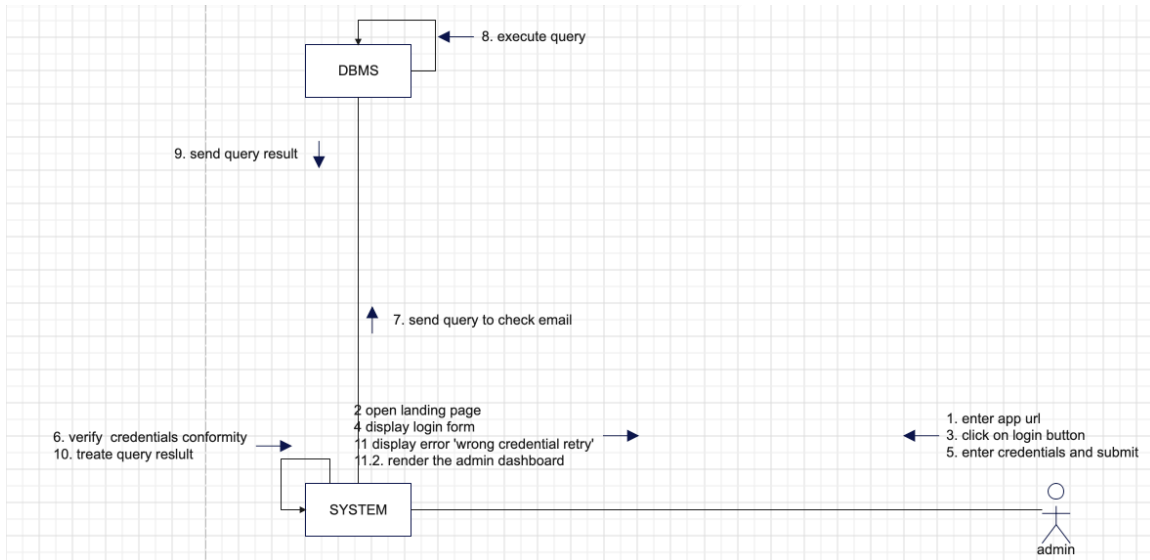


Figure 13 :Communication diagram of authentication

v. *Communication Diagram <<create user>>*

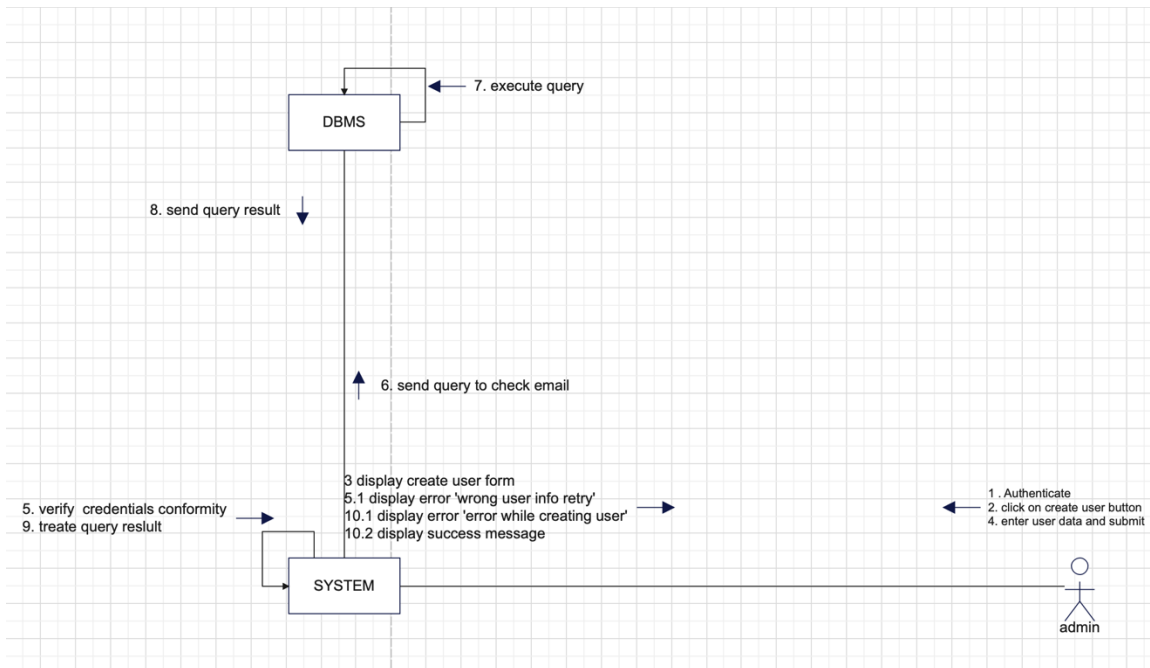


Figure 14 Communication Diagram create user

vi. *Communication Diagram <<Send Availability >>*

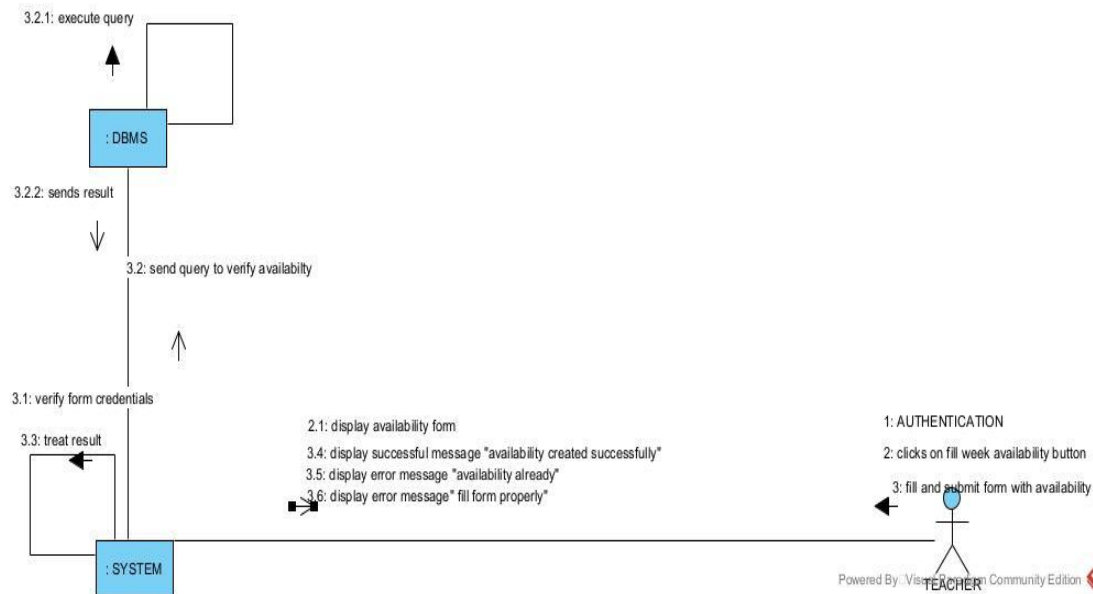


Figure 15 : Communication diagram of set availability

XI. 2.3. SEQUENCE DIAGRAM

i. Definition

A sequence diagram is a UML diagram that illustrates the sequence of messages between objects in an interaction. It consists of a group of objects that are presented by lifelines and the messages that they exchange over time during the interaction. These interactions are arranged from top to bottom following their order of occurrence.

ii. Formalism

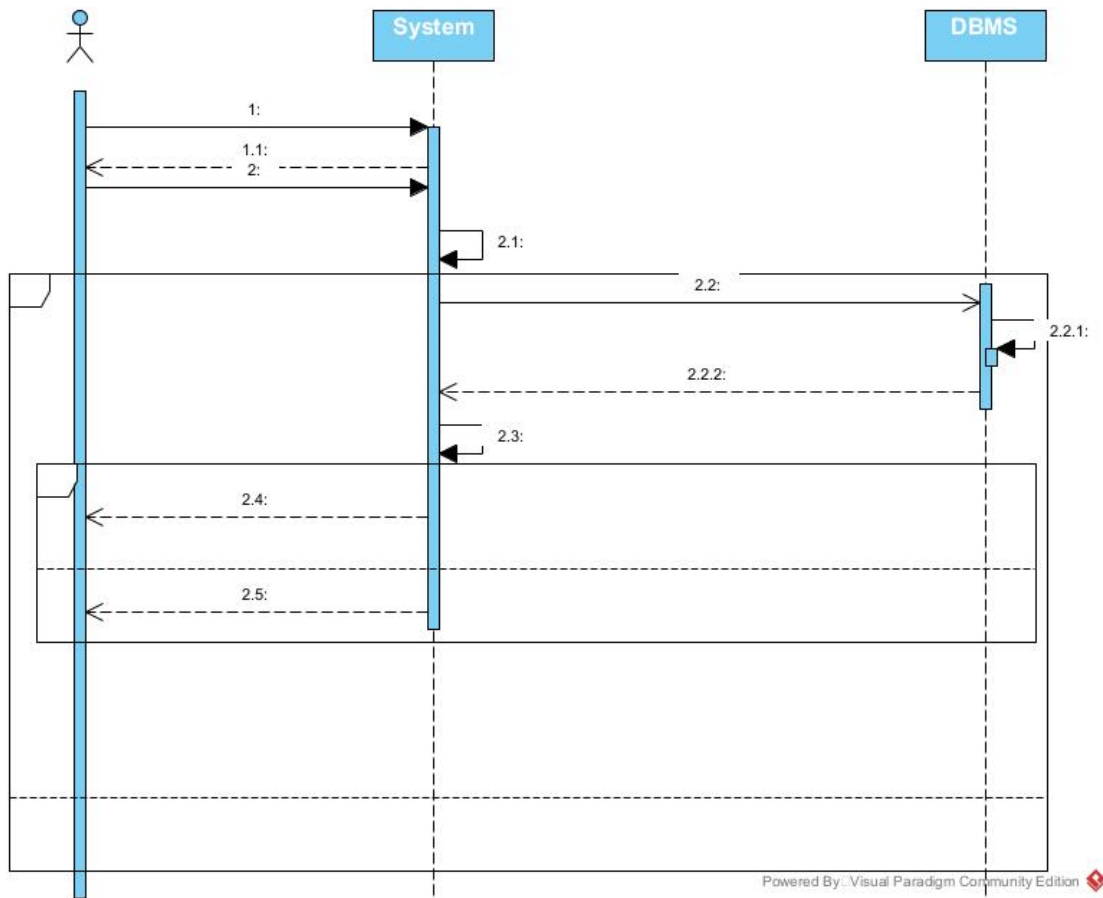
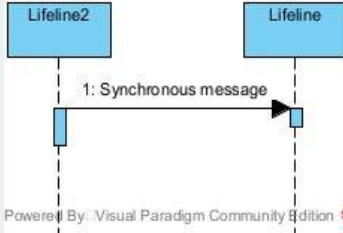
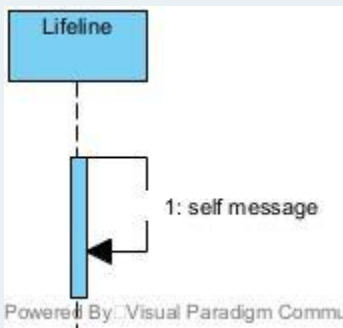
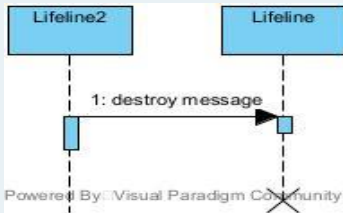
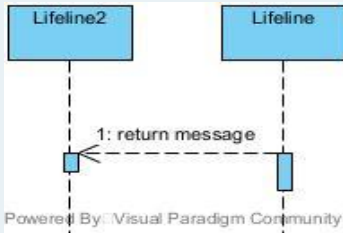


Figure 16:Sequence Diagram formalism

iii. Components of a Sequence Diagram

Table 14:Components of a Sequence Diagram

Element	Description and main properties	Representation
Lifeline	They represent rows or objects instances that participates in sequence being modeled.	

Synchronous Message	A message that represented an invocation of an operation of the target lifeline	 Powered By: Visual Paradigm Community Edition
Self Message	Represents the execution operation call on the same object	 Powered By: Visual Paradigm Community Edition
Destroy Message	It destroys a participant in the course of a process	 Powered By: Visual Paradigm Community Edition
Return Message	Represents a response to a message	 Powered By: Visual Paradigm Community Edition

iv. Sequence Diagram <<Authentication>>

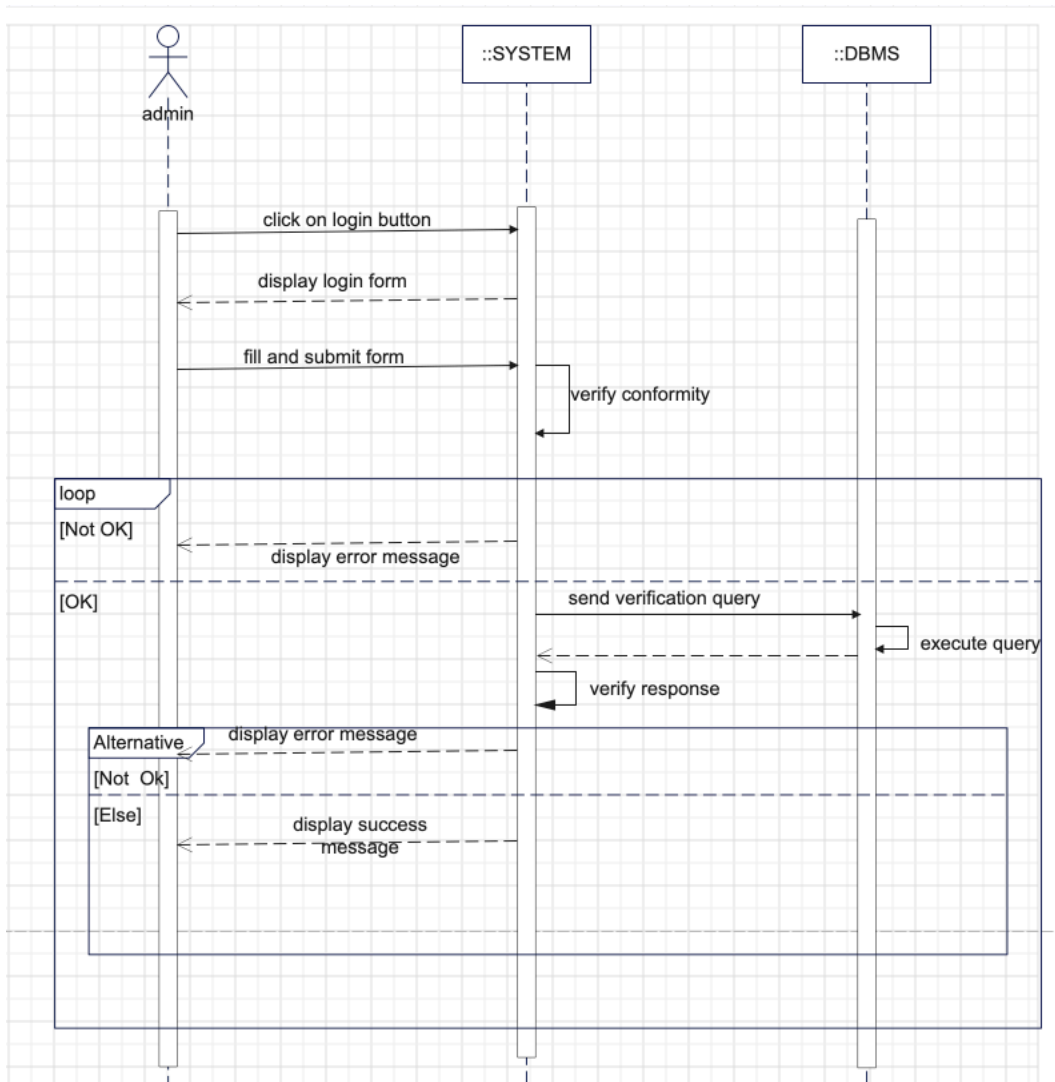


Figure 17: Sequence Diagram Authentication

v. Sequence Diagram <<Create user>>

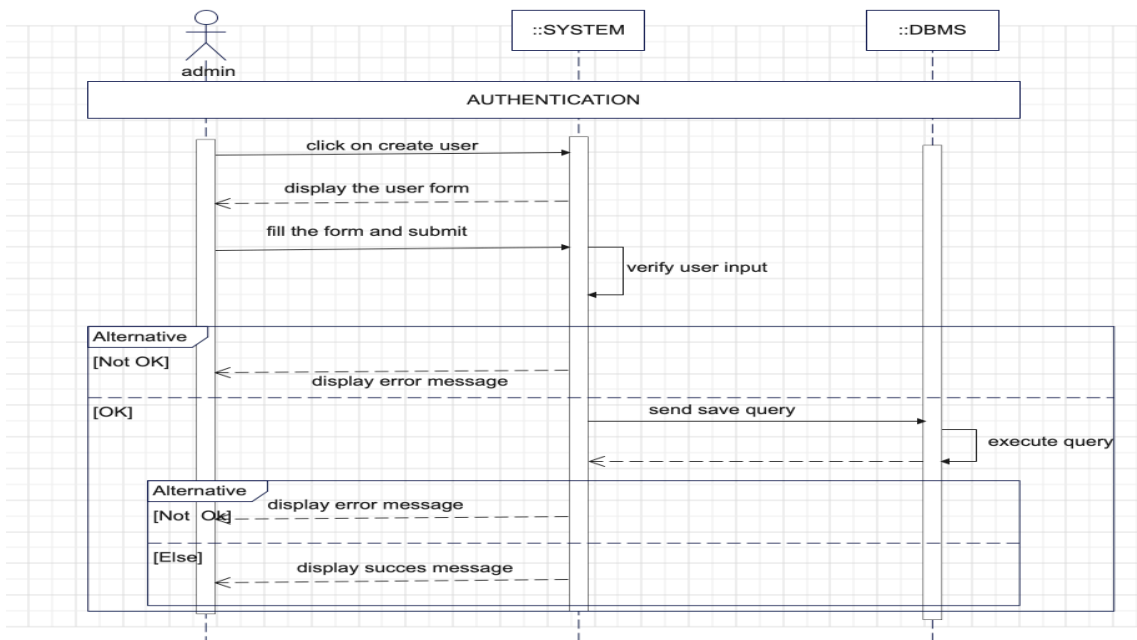


Figure 18: sequence diagram create user

vi sequence Diagram <<send Availability >>

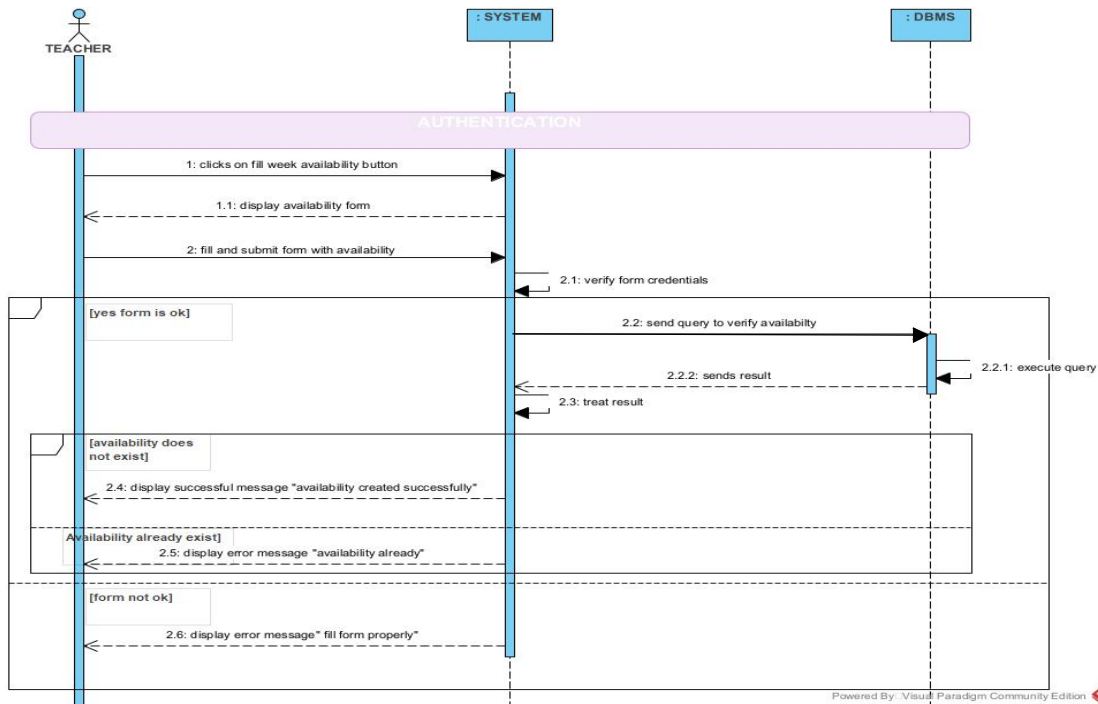


Figure 19: sequence diagram of send availability

XII. 2.4 ACTIVITY DIAGRAM

i. Definition

An activity diagram provides a view of the behaviour of a system by describing a sequence of actions in the process. The activity diagram makes it possible to emphasize the processing and to graphically formalize the sequence of actors carried out in a use case.

ii. Formalism

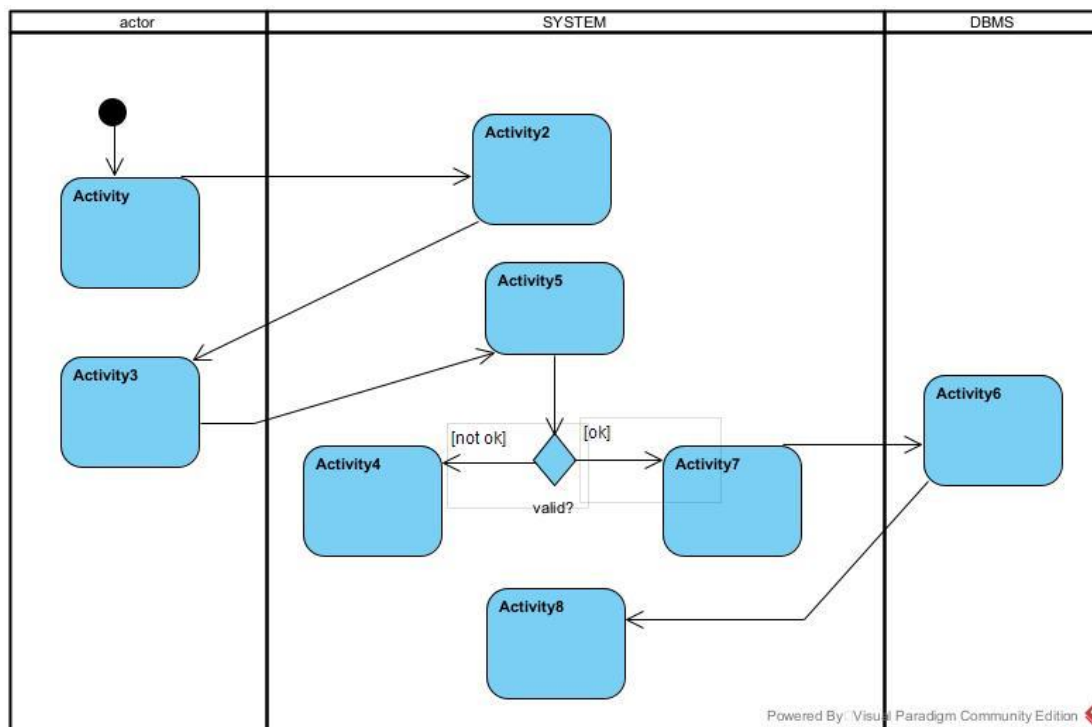
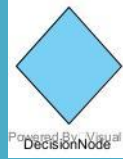
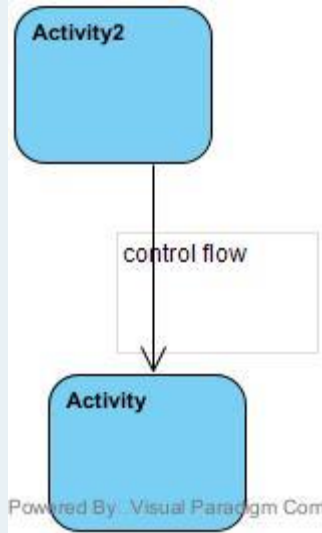


Figure 20: activity diagram formalism

iii. Components of an Activity Diagram

Table 15: Component of an activity diagram

Decision node	It is a decision node that takes a condition from one edge and presents the different options or possibilities to cross the outgoing edges. It is generally represented by a diamond	
Initial Node	It is a control node at which flow starts when an activity is invoked	
Final Node	A control node that stops all flow in an activity. It is represented by a circle with a filled circle inside	
Control Flow	An edge that starts and activates a node after the previous one is finished	

Action	Represents a single step in activity which cannot be further decomposed	
--------	--	---

iv. Activity Diagram<< Authentication >>

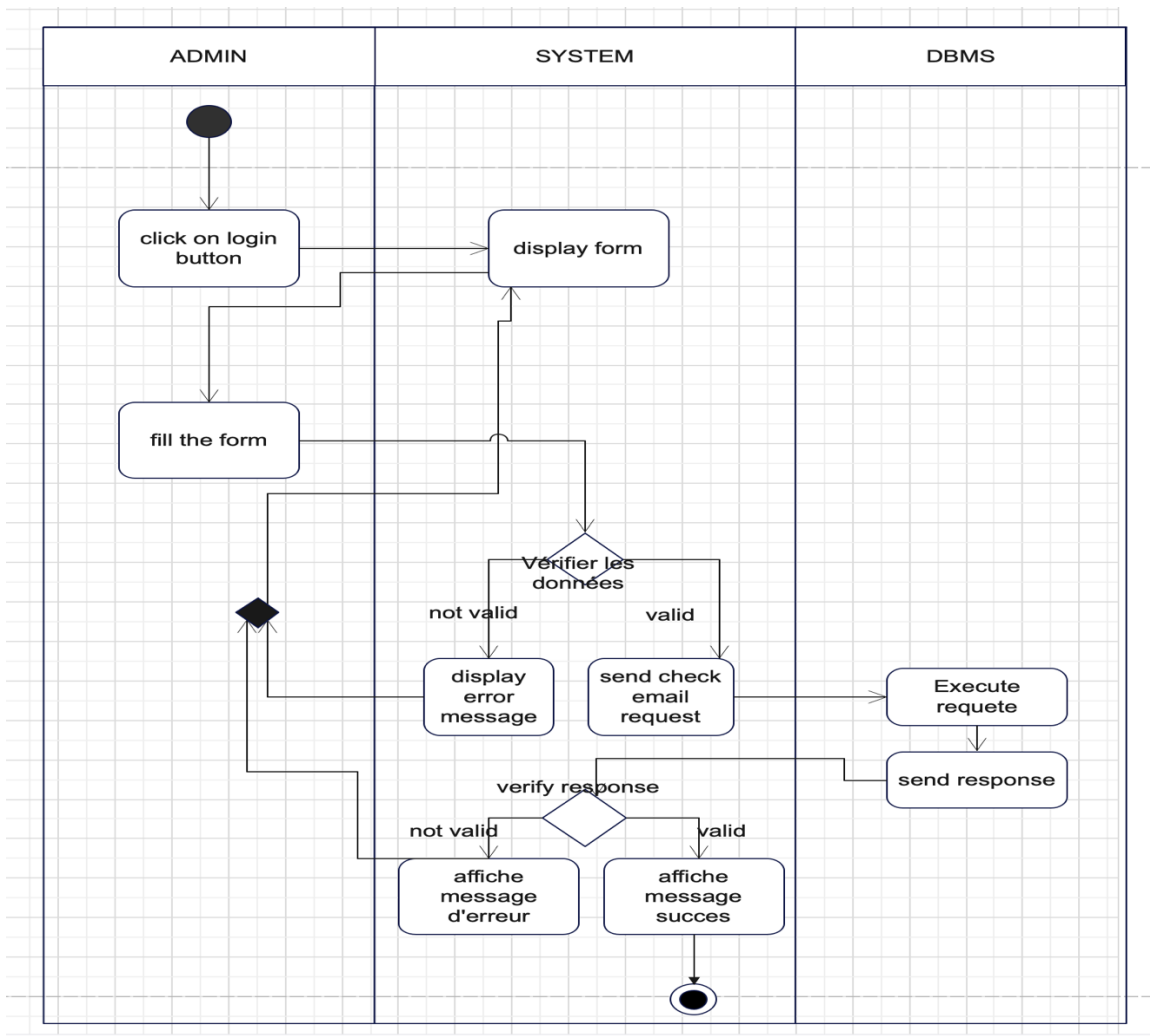


Figure 21 activity diagram authentication

iv. Activity Diagram<< create user>>

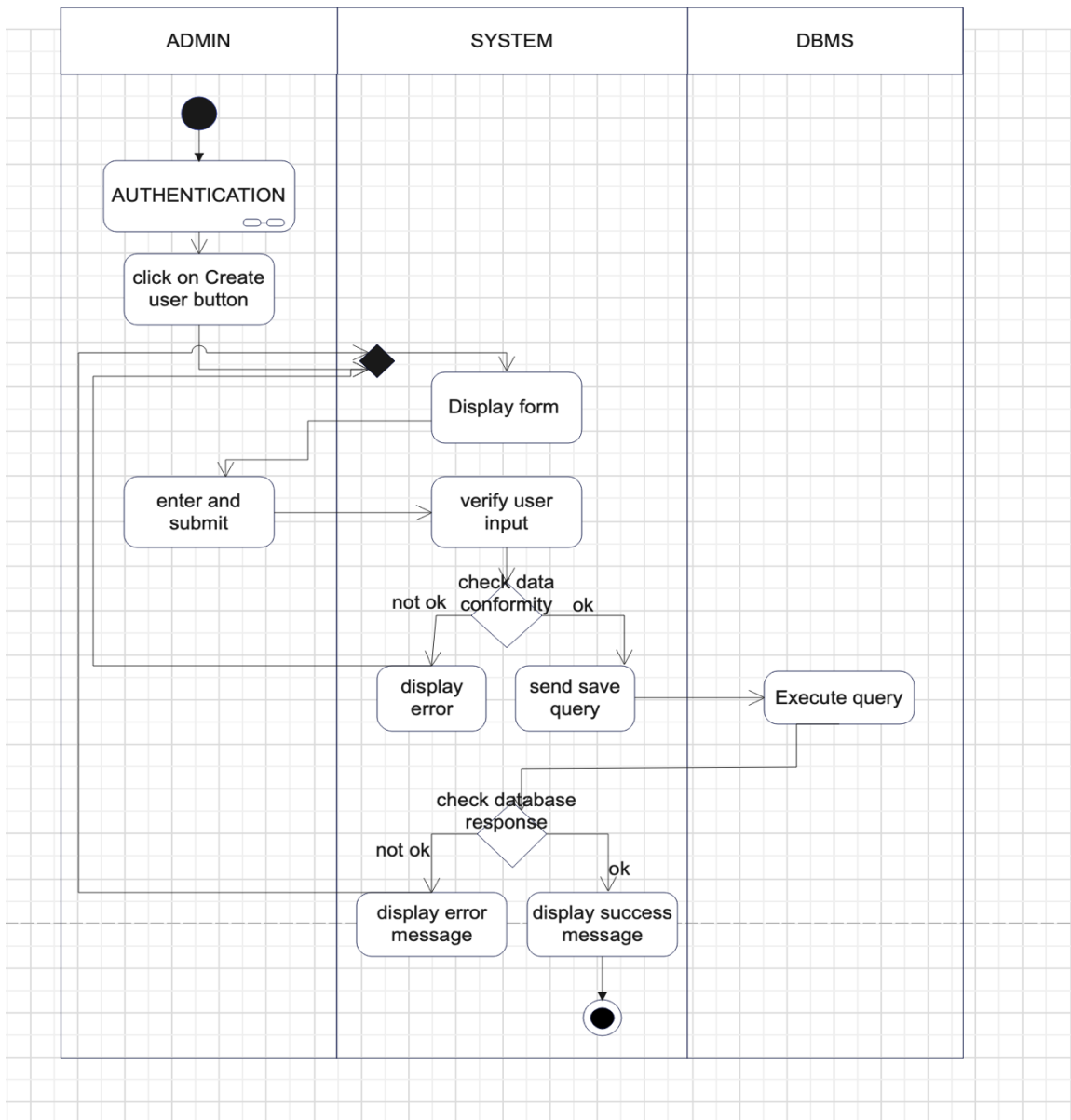


Figure 22: create user activity diagram

v. Activity Diagram<<send Availability >>

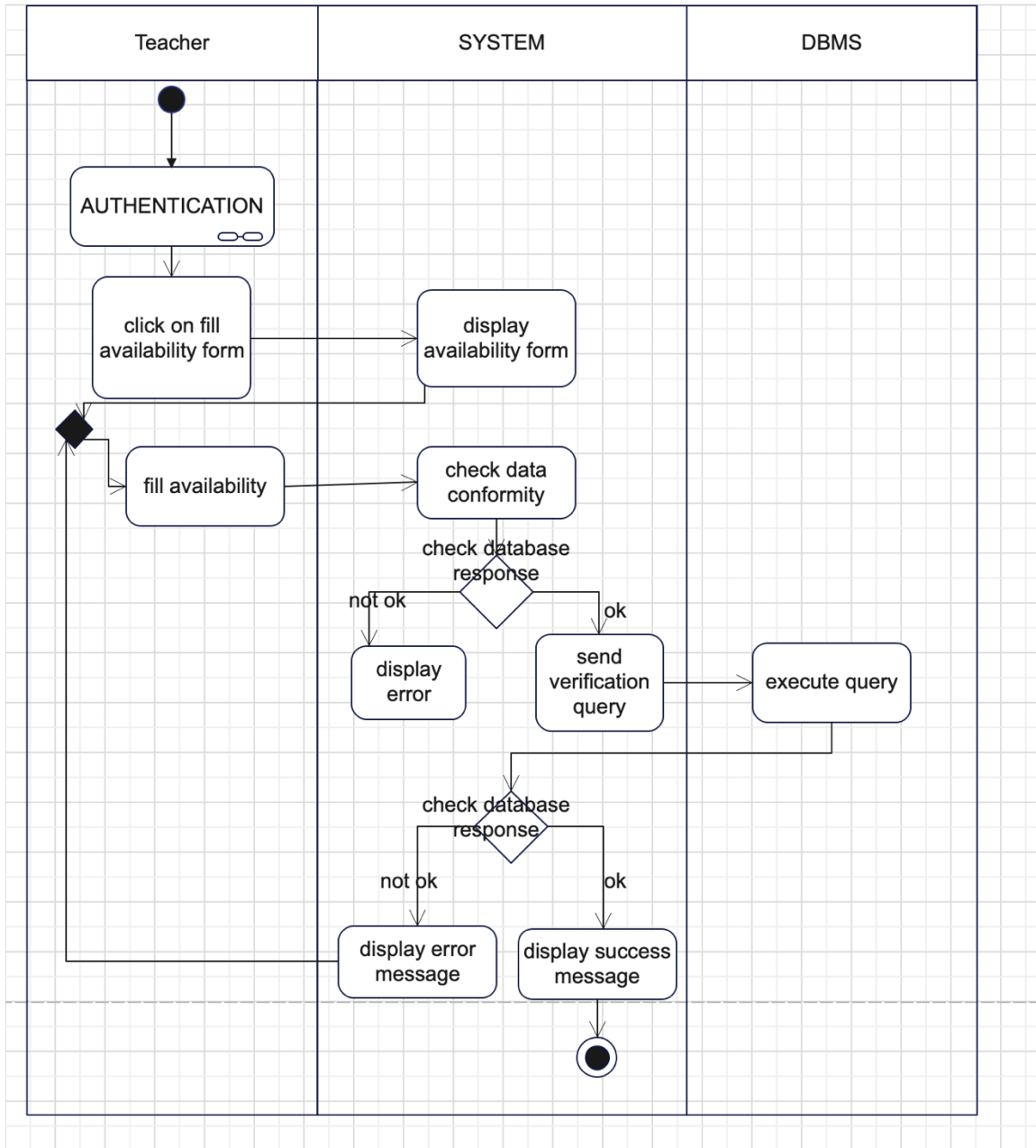


Figure 23 : Activity diagram of set availability



CONCLUSION

Having reached the end of our analysis, we were asked to illustrate the language and method used during this phase, present the different diagrams of the functional branch. we can then say that this analysis file has allowed us delimit the functional needs of the web application that will be designed and to have a detailed overview of the new system to be set up without notwithstanding we quickly dive into the conception phase

CHAPTER V: CONCEPTION PHASE

INTRODUCTION

I. CLASS DIAGRAMME

II. STATE MACHINE DIAGRAM.

III. PACKAGE DIAGRAM

CONCLUSION



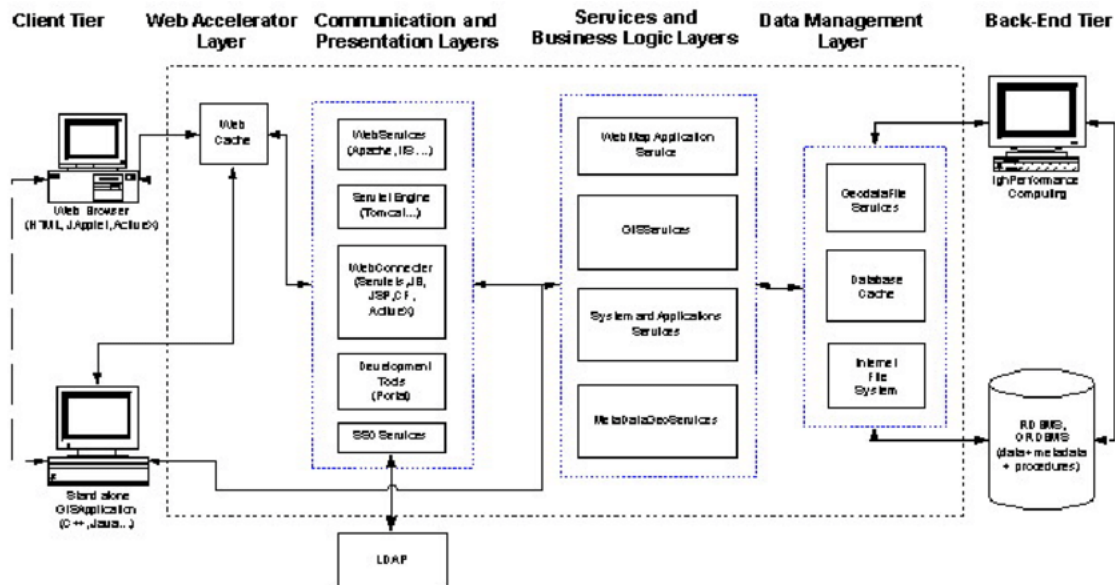
INTRODUCTION

The realization of an analysis file is very important for the advancement of the application; it guarantees the reliability of maintenance and use of a software. That said, after analysing our system, we will move on to the design of our future system. To do this we will follow the steps of the technical branch of the 2TUP process, making the diagrams associated with these steps namely the class diagram and the state machine diagram.

a. A. CAPTURE OF TECHNICAL NEEDS

PHYSICAL ARCHITECTURE

The design of the database heavily relies on its underlying architecture. In this project, we utilized an n-tier architecture, a software design pattern that partitions the entire system into multiple, related but separate layers or tiers. This separation allows for greater flexibility, scalability, and maintainability, as each tier can be independently modified, altered, changed, or replaced without affecting the entire system.



Source: <https://www.mindphp.com/images>

Figure 24: System physical architecture

LOGICAL ARCHITECTURE

The application logic of our system is based on MVC design pattern which is a very practical way to organize the different classes in an application development project facilitating error detection, collaboration, and code reusability.

MVC (Model View Controller) design pattern is an architecture and a design method that organizes the human-machine interface (HMI), a view (user Interface) and a controller (Control Logic, event management, synchronization), each having a specific role in the interface.

This design pattern allows you to organize your code well, its goal is to separate you code into three main parts and in separate files namely: **Model, View, Controller**

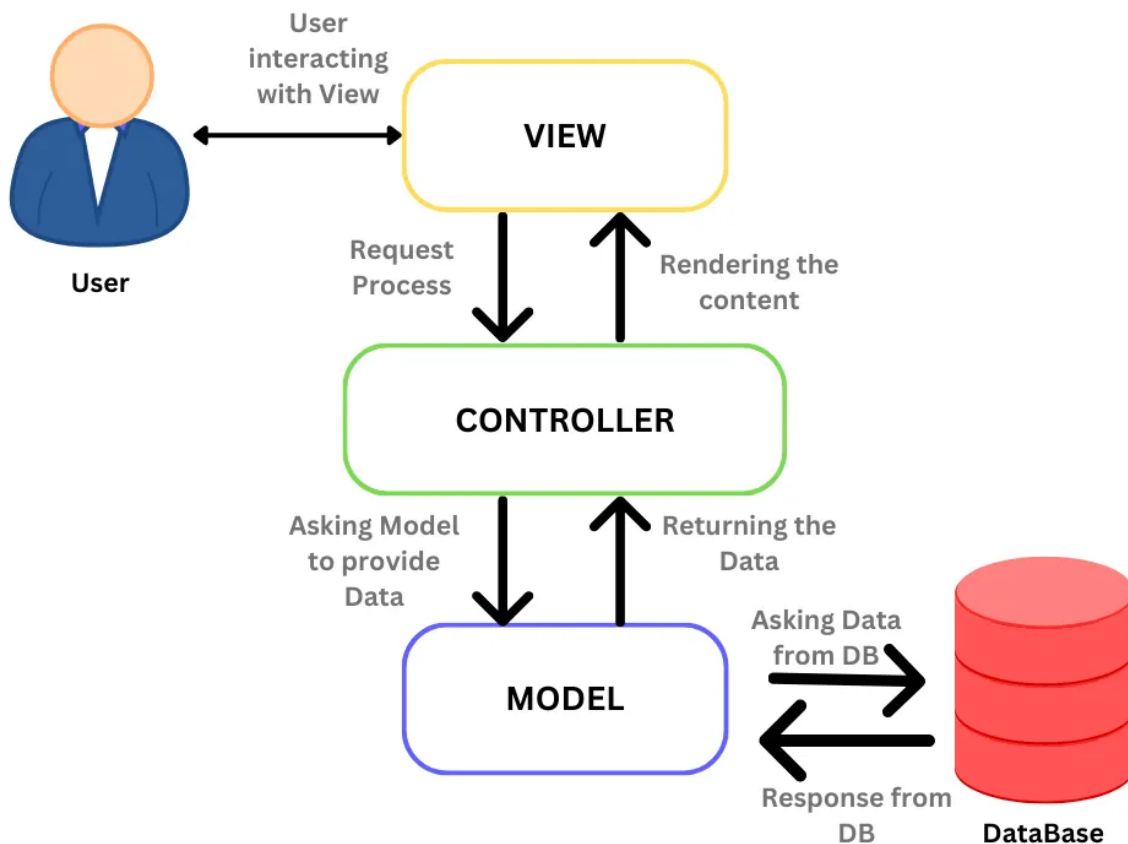


Figure 25: MVC illustration diagram

Source: <https://www.mindphp.com/images/knowledge/IOT/ASP.NET-MVC.JPG>

b. I. CLASS DIAGRAM

Definition

The class diagram expresses the static structure of the system in terms of classes and the relationships between those classes. The interest of the class diagram is to model the entities of the information system. The class diagram is used to represent all the finalized information that is managed by the domain. This information is being structured, that is, it has been grouped into classes.

XIII. ii. Formalism

The diagram highlights possible relationships between these classes. A class is a set of elements of the same nature that can be uniquely identified. This diagram has several concepts: class, attribute, identifier, method, access modifier. (Public (+), protected (#), private (-), package ()), derivative (/), relationships and crossmultiplicities. The table below shows the different components of a class.

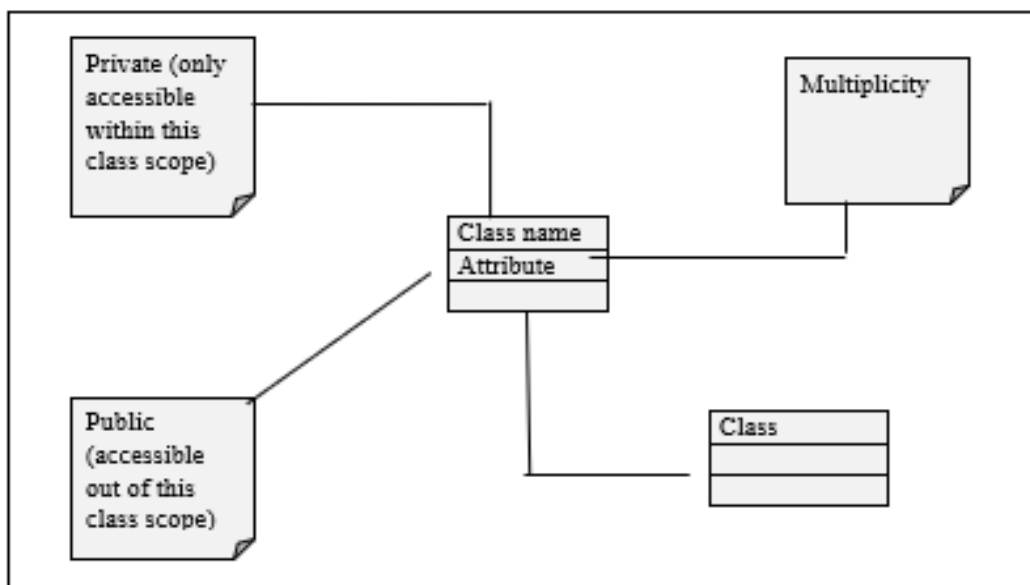
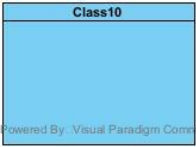
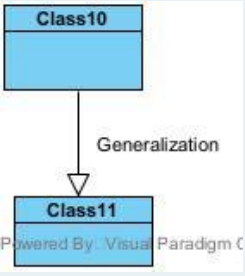
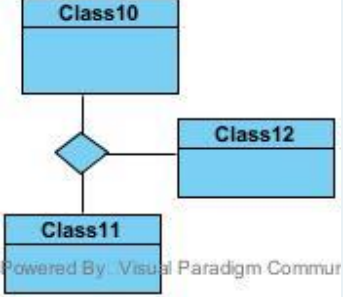
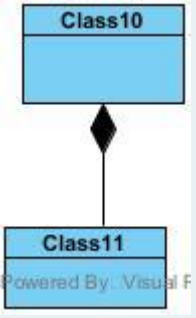


Figure 26: Class diagram formalism

Components of a Class Diagram

Table 16 : Component of a class diagram

Element	Description and main features	Representation
Class	Represents an entity within a system. It is an abstract representation or description of a set of objects in the application domain	
Generalization	Exist between two related classes where one is a specialized form of the other.	
N-ary relation	It is a type of relationship used to model several classes	
Composite Association	It is a type of association that represents a part-whole relationship in which one element dies too when the other dies. It is represented by a black diamond	

Graphical Representation of System Class Diagram

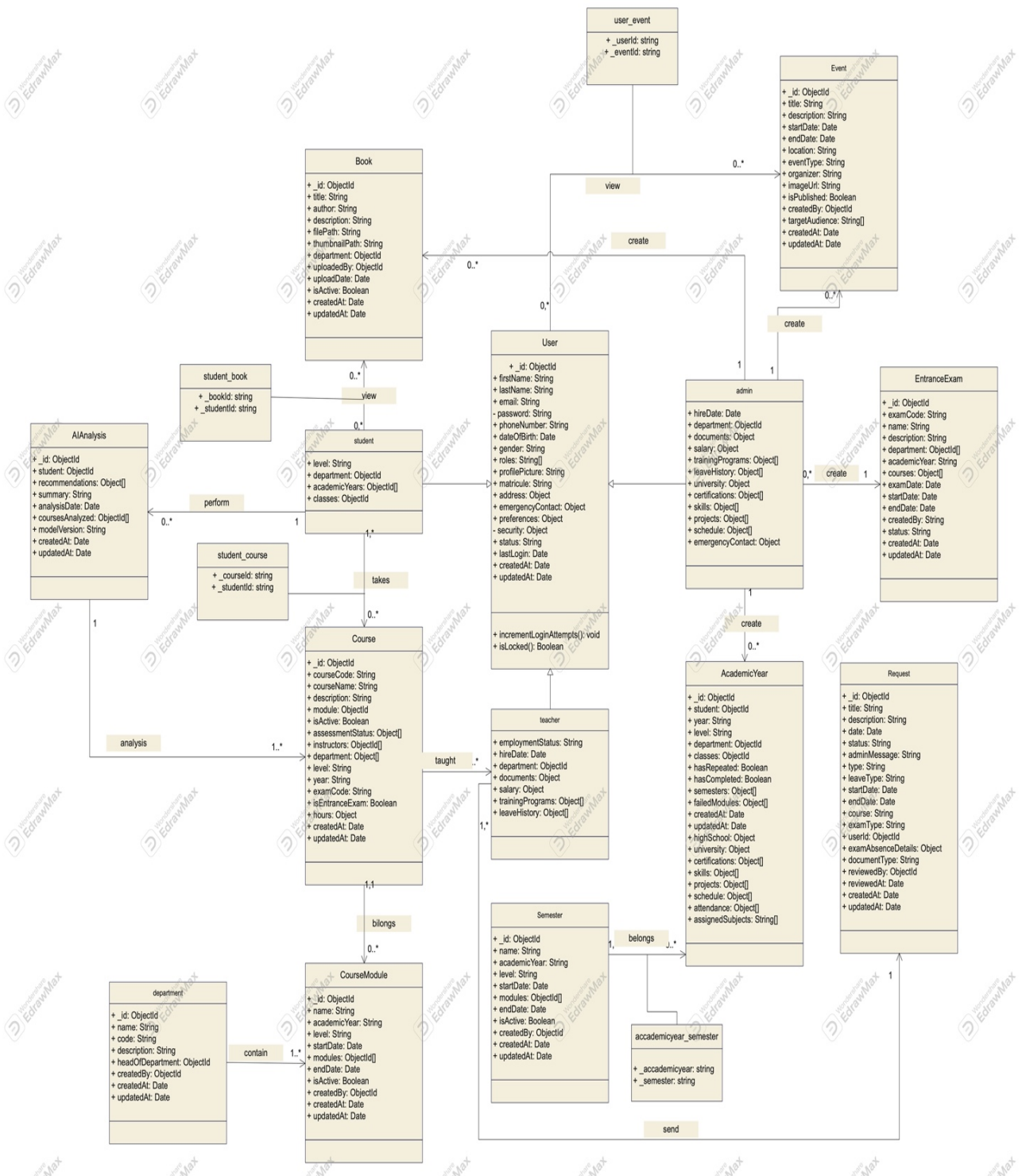


Figure 27: System class diagram

a. II. STATE MACHINE DIAGRAM

Definition

State diagrams are used to document the various modes (states) that a class can take, and the events that cause a state transition. Their role is to present the treatments (operations) that will manage the field studied. They define the sequence of class states, also known as the state chart diagram, which models the dynamic flow of control from the state of a particular object within a system.

Formalism

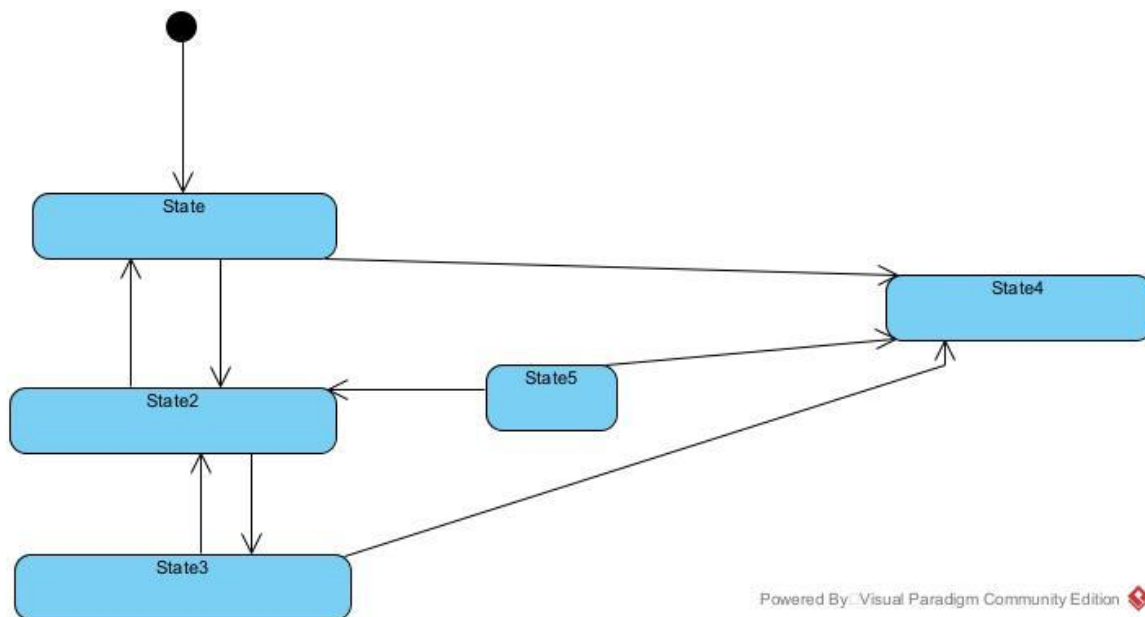
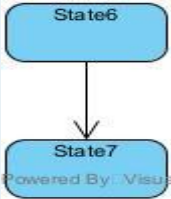


Figure 28: State machine formalism

Components of a State Machine Diagram

Table 17 : Component of state machine diagram

Element	Description and features	Representation
Initial State	It represents the final or beginning of state and it is represented by a block filled circle	
Final State	It represents the final state or end of a system. It is represented by a filled circle within another circle	
Transition	it is a change of control from one state to another due to the occurrence of some events. It is represented by an arrow labelled with the event that causes it's change of state.	
State Box	It depicts the condition or circumstances of particular object of a class at a specific point of time.	

State Machine <<Payment>>

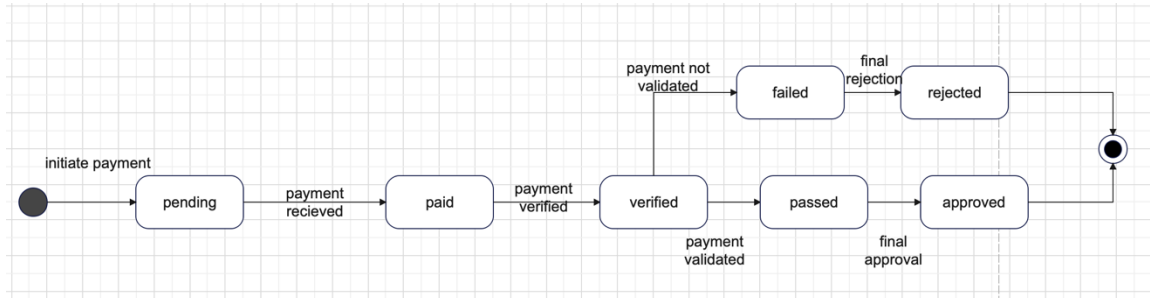
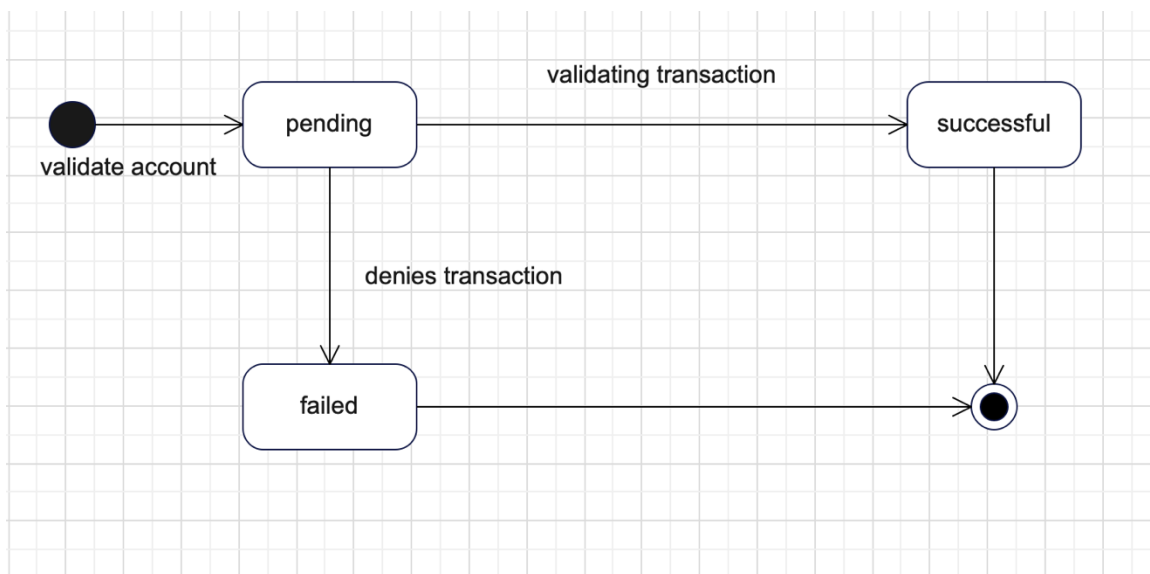


Figure 29 Payment state machine

State Machine <<State machine Validate account>>



b. III. PACKAGE DIAGRAM

Definition

They are structural diagrams used to show the organization and arrangement of various model elements in the form of a package. A package diagram is group of related UML elements such as diagrams, documents, classes, or even other packages. Package diagrams are commonly used to provide a visual organization of the architectural layout within any UML classifier such as software systems.

Formalism

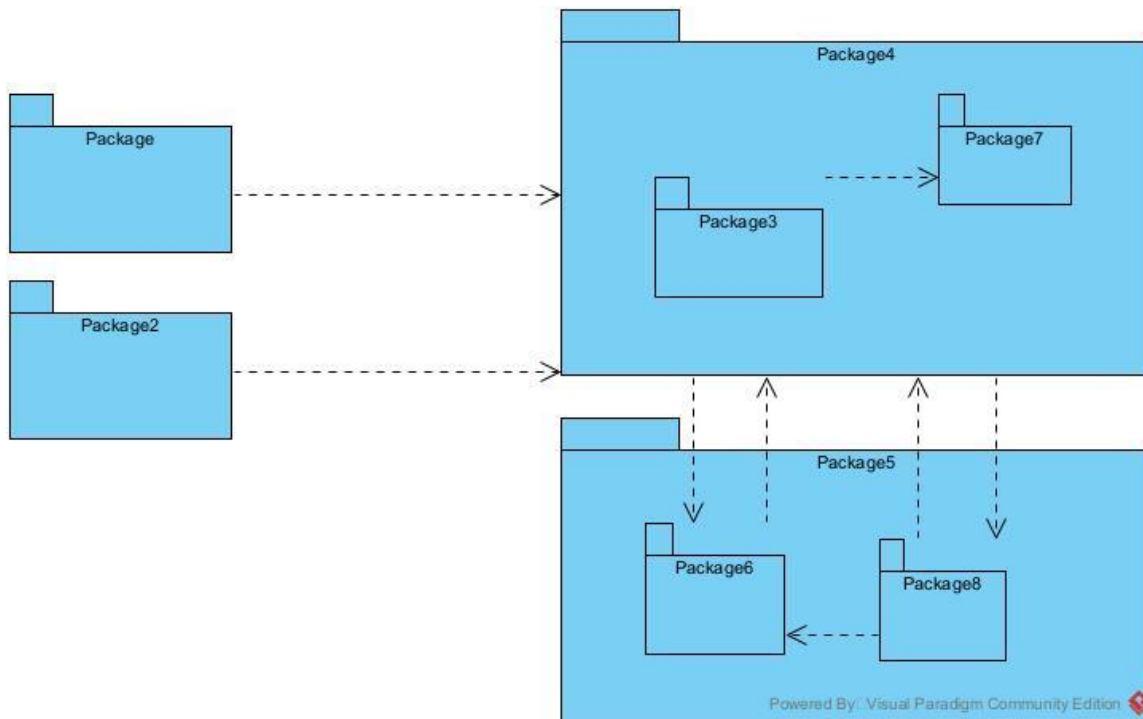
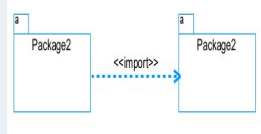
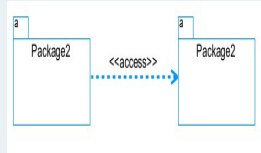
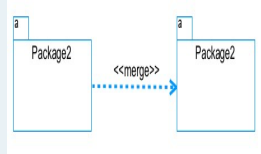


Figure 30:Package diagram formalism

Component of package diagram

Table 18: Component package diagram

Element	Description and main features	Representation
Package	It is namespace used to group together logically related elements within a system. Each element contains within the package should be a packageable element and have a unique name	
Package import	A relationship Indicate that, functionality has been imported from one package to another.	
Package access	A relationship Indicates that one package requires assistance from the function of another package.	
Package merge	It is a relationship which shows that, the functionality of two packages are combines to a single function.	

System package diagram

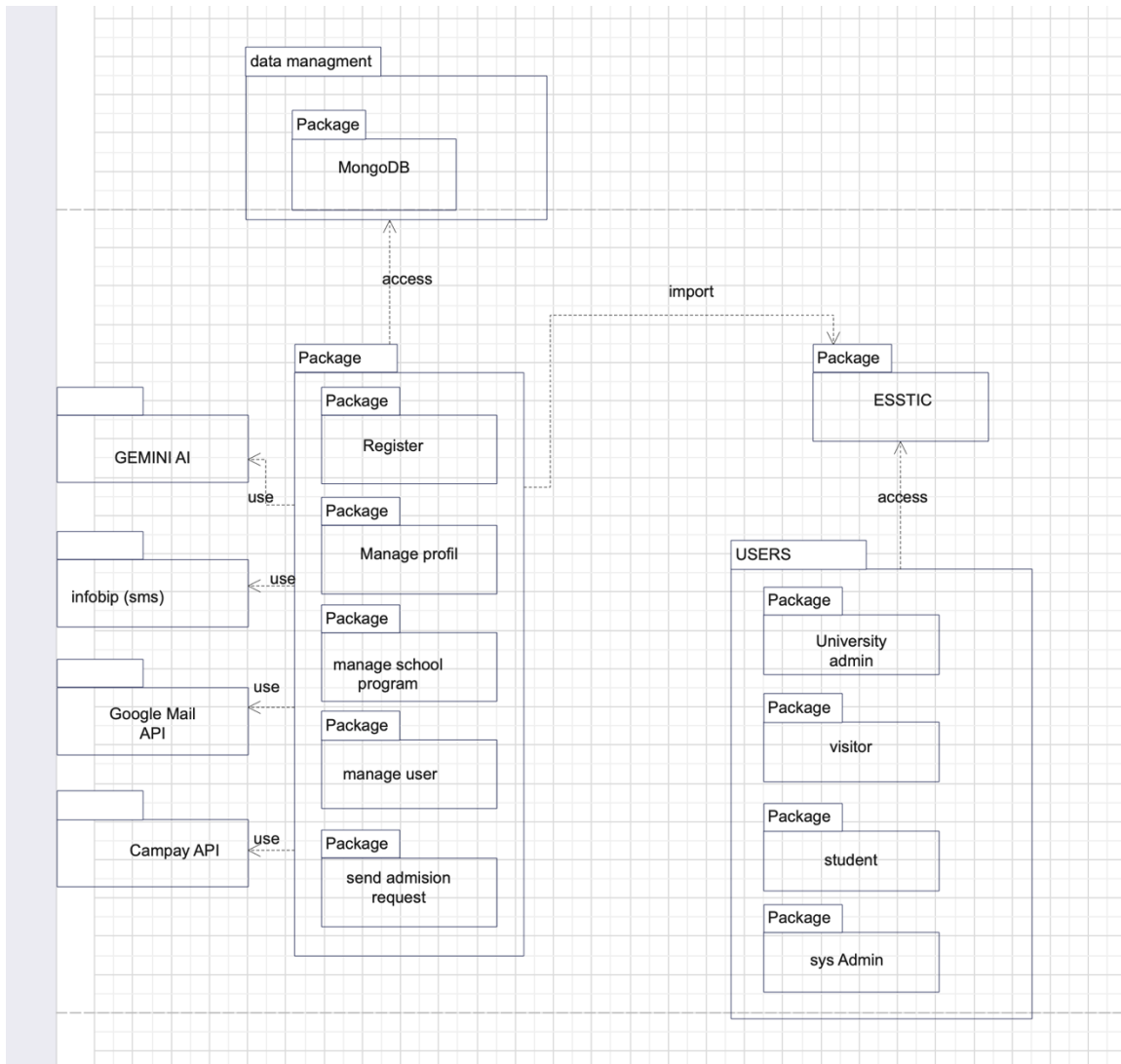


Figure 31: System package diagram



CONCLUSION

The conception phase allowed us to bring out the data necessary for the creation of our database and the implementation of our application. The different elements modeled in this part allowed us to have a global view of the different modules of our application; therefore, the next step of our project will be the drafting of the realization file taking into account the different elements modelled above.



CHAPTER VI: REALIZATION PHASE

Preamble

The realization document consists in the implementation of the project in a programming language according to the specifications defined in the previous file. It contains component and deployment diagrams; the testing and development phases. At the end of this part, programming documentation will be produced explaining the architecture of the database and the architecture of our code.

Content :

INTRODUCTION

I. PRELIMINARY CONCEPTION

1. DEPLOYEMENT DIAGRAM

2. COMPONENT DIAGRAM

II. TECHNOLOGICAL CHOICES

1. Material Choices

2. Software Choices

3. Technological Choices

III. TECHNOLOGICAL CHOICES

4. Material Choices

5. Software Choices

6. Technological Choices

CONCLUSION



INTRODUCTION

Design and modelling are important links in the development process of an application because they make it possible to define and specify the different elements constituting the application or system studied. Realization is the culmination of the work carried out upstream because it makes it possible to physically create the various objects that were designed, and to bring the computer system to life. Throughout this phase, we will present the software, physical and hardware architecture of our system, through diagrams such as the component diagram, deployment diagram, physical model and code architecture.

1. DEPLOYMENT DIAGRAM

i. Definition

The deployment diagram models the hardware components used to implement a system and the association between those components. Deployment diagrams can be implemented as early as the design phase to document the physical architecture of the system. As elements used in deployment diagrams we can have, components, as in component diagrams, and nodes, which represent the physical processing resources of the system, and their associations.

ii.) Formalism

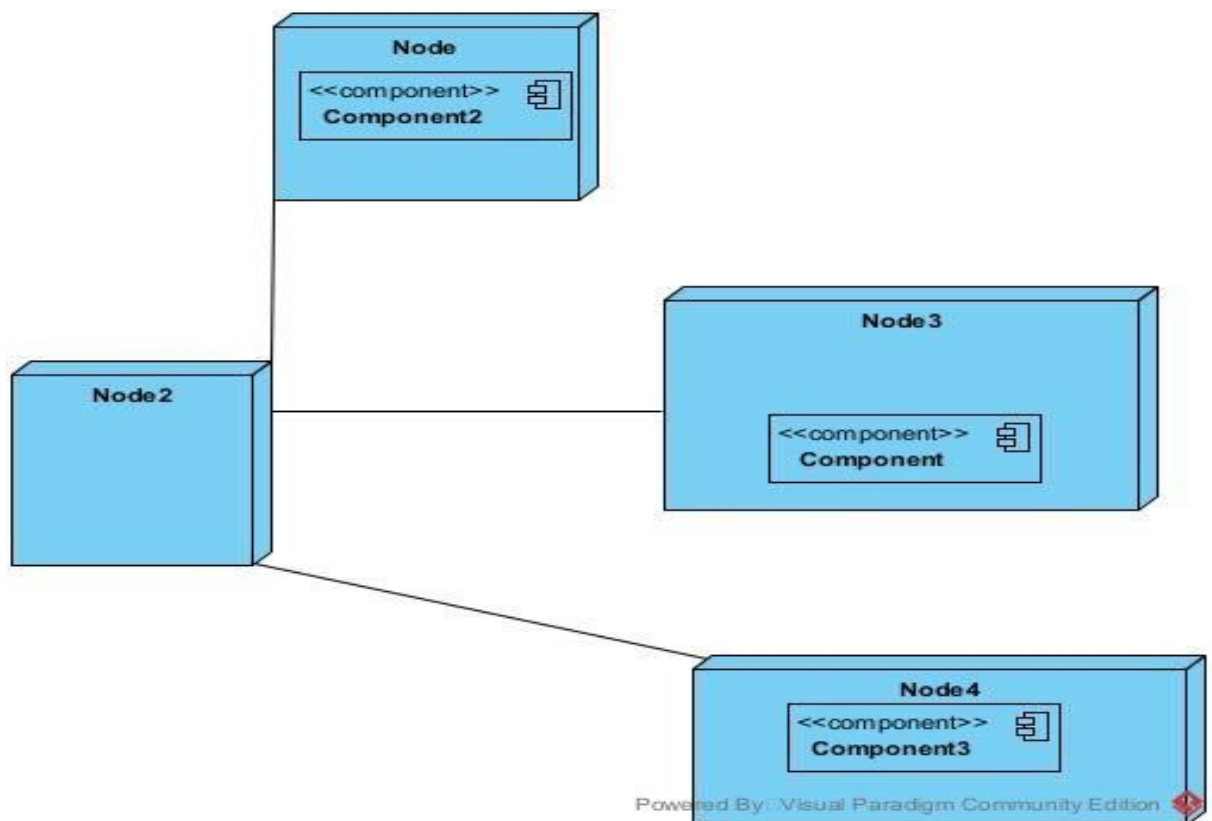
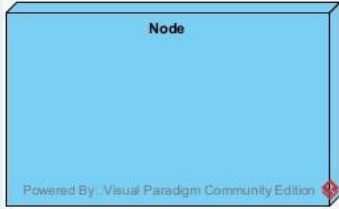
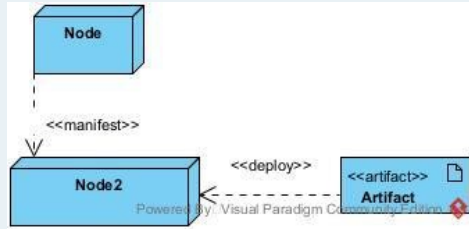


Figure 32: Component diagram formalism

Components of a Deployment Diagram

Table 19: Component of a deployment diagram

Element	Description and main	Representation features
Component	A component represents a software entity of the system	
Node	A node represents a set of hardware elements of the system. This entity or node is represented by a three dimensional cube	
Interface	Represents a declaration of a set of coherent public features and obligations	
Dependency (manifestation deployment)	A relationship between two model elements where changes in one of them will similarly affect the other	

XIV. iv. Deployment diagram of ESSTIC

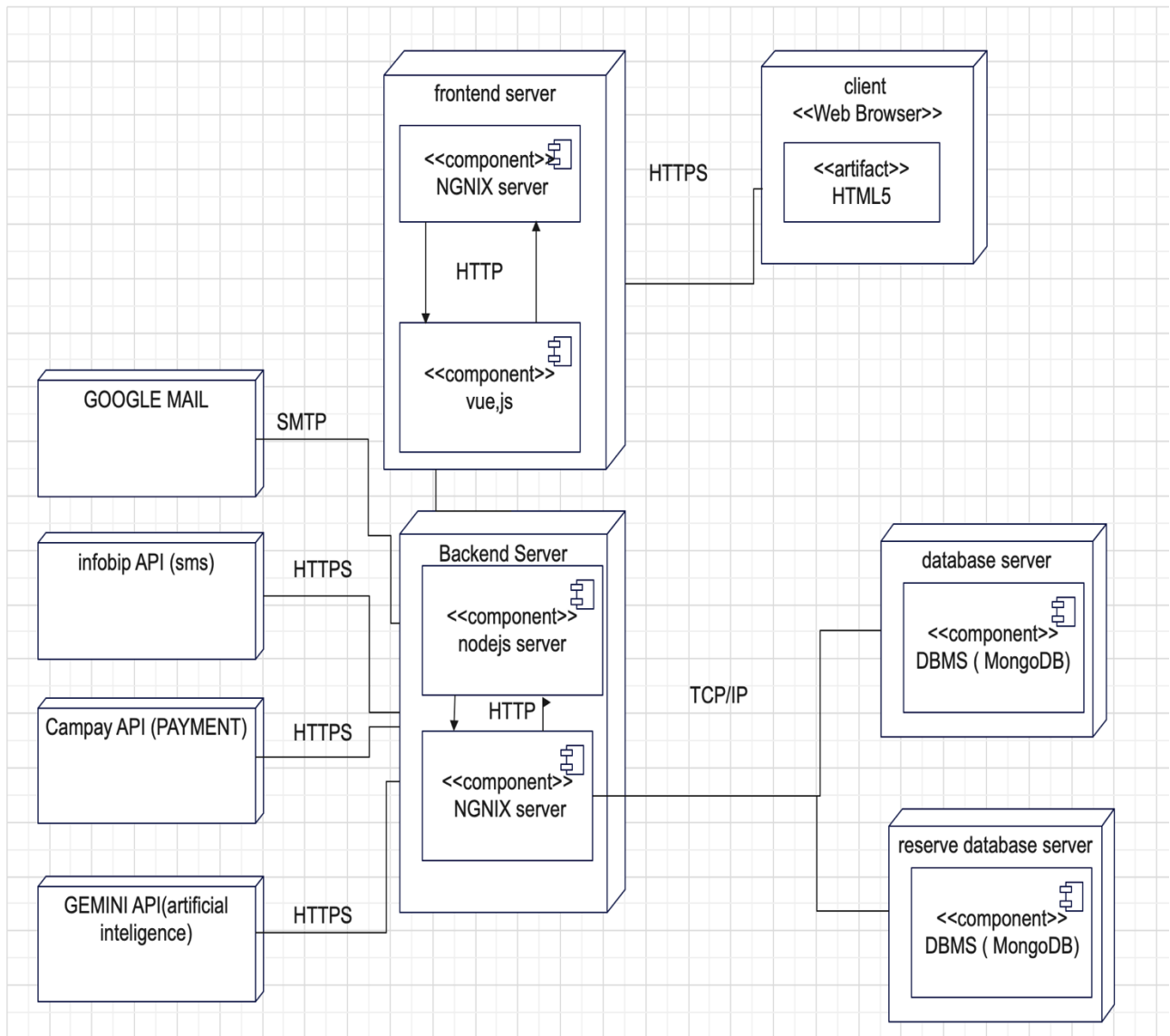


Figure 33 : System deployment diagram

COMPONENT DIAGRAM

1. Definition

The component diagram is a UML diagram that provides a graphical representation of dependencies and generalizations between software components, including source code components, binary code components, and executable components.

2. Formalism

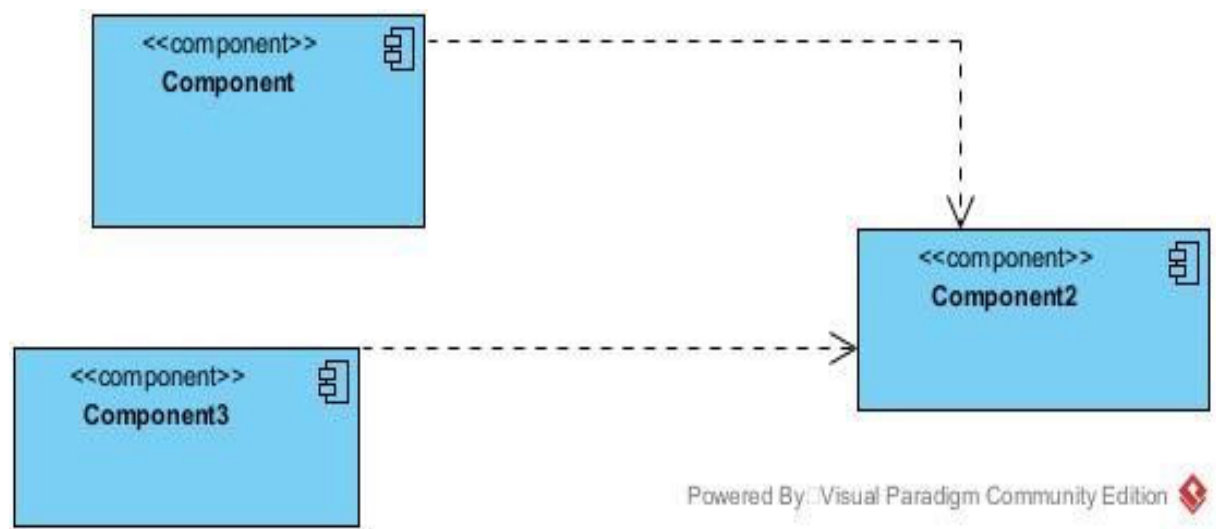
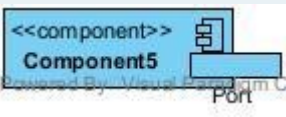
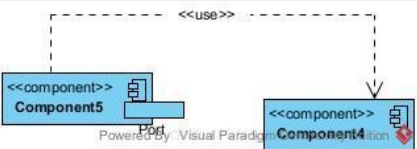


Figure 34: Component diagram formalism

Components of a Component Diagram

Table 20: Component of a component diagram

Element	Description and main features	Representation
Component	Represents a modular part of system that encapsulates its content. In other words it is a part of a system or application.	
Interface	Represents a declaration of a set of coherent public features and obligations.	
Part	Specifies a distinct point between that classifier element and its environment.	
Usage	A relationship in which one element requires another element for its full implementation operation.	



CONCEPTION AND IMPLEMENTATION OF AN INTELLIGENT SCHOOL MANAGEMENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND LEARNING ORIENTATION (CASE STUDY: ESSTIC)



Instance Specification	It has the capability of being a deployment target in a deployment relationship	
---------------------------	--	---

3. Component Diagram of ESSTIC

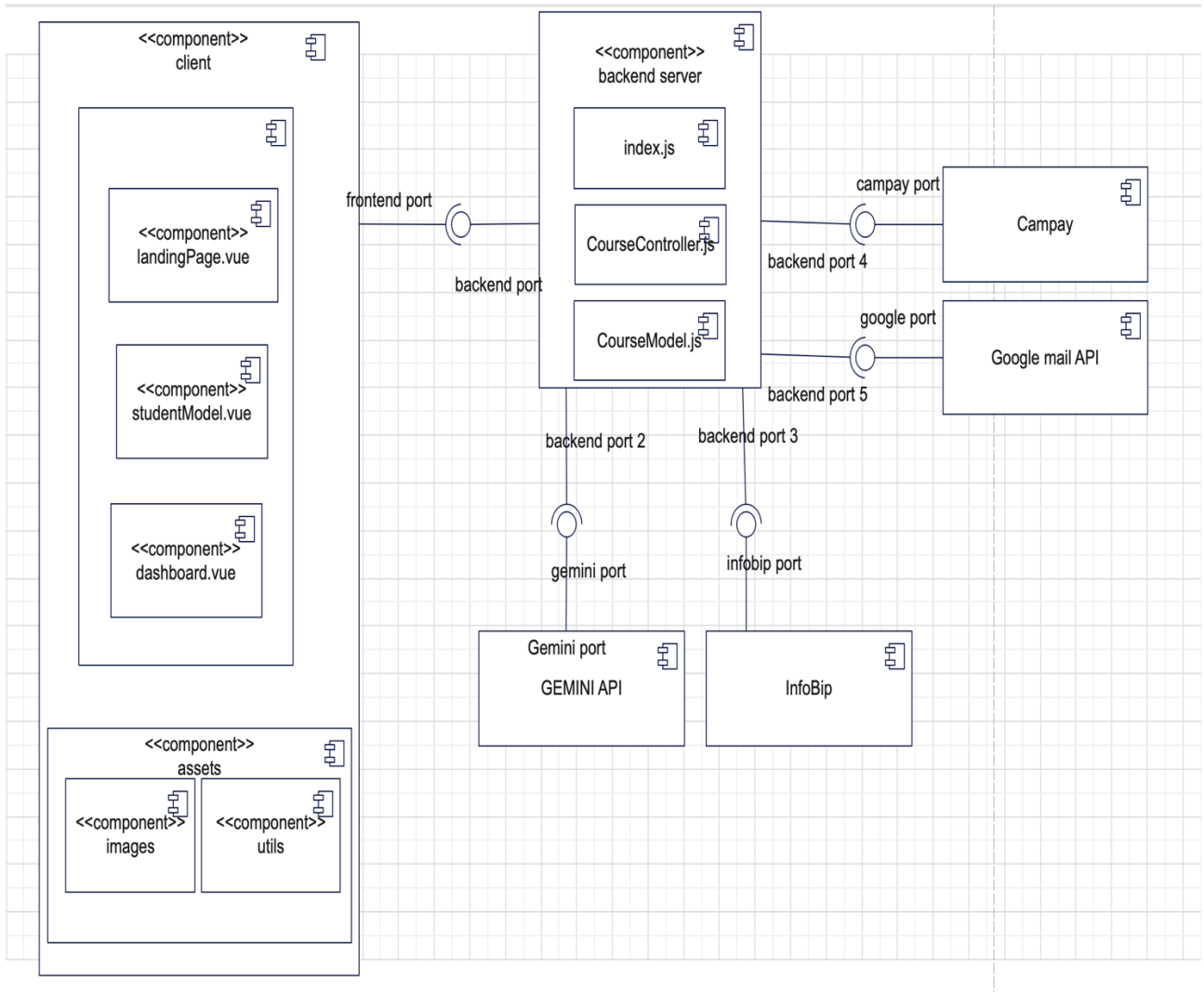


Figure 35 : Component Diagram of ESSTIC

b. TECHNICAL CHOICES

A. Materials used for the development of the ESSTIC Smart System

i. Hardware Material

Table 21: Hardware materials used for the development of the ESSTIC Smart System

Material	Quantity	Function	Image
Dell LATITUDE E7540	1	The computer was used for the programming of the micro controller, the analysis, research, the design of the system architecture. This was the main component in this project.	
HP computer	1	The computer was used for the development of the mobile application and web application	
iPhone 11 pro max	1	Testing the mobile app functionalities and debugging.	

Table 22: Software materials used for the development of the ESSTIC Smart System

Software	Role	Reason for use	Image
OS WINDOWS 10	The operating system required to run the software is the one on which we have worked.	Windows 10 has a user-friendly interface that is familiar to many users, making it easy to navigate and use.	
Visual Paradigm	The software engineering workshop used for solution modelling. We utilized this tool for modelling the various diagrams of our system	. Supports multiple modelling languages and methodologies, making it suitable for various project requirements and enabling teams to adapt to different development processes.	
Visual studio code	Visual Studio Code is a lightweight, open-source code editor developed by Microsoft, supporting multiple programming languages. Using for writing of application code.	. Combines tools for version control, debugging, and terminal access in one interface, improving efficiency and reducing the need for multiple applications.	
Github	It is a website and cloud service that helps developers store and manage their code.	Git allows developers to track changes, revert to previous versions, and maintain a history of modifications, ensuring that	

		no work is lost and enabling easy recovery.	
Flutter Framework	A popular open-source framework for building natively compiled, cross-platform applications. Provides a widget-based approach for efficient rendering and state management	Promotes the creation of reusable components, which enhances code maintainability and allows for easier updates and testing.	
Node.js	Node.js is an open-source and cross-platform JavaScript runtime environment. It is a popular tool for almost any kind of project!	It is used to build our backend, helping us to have a dynamic web application	
MySQL	MySQL is a popular open-source relational database management system	It is used to store and manage our data	
MS Excel	MS Excel is a powerful spreadsheet application used for data organization, analysis, and visualization. Using to create charts after the	Provides a variety of charting and graphing options, helping users to present data in a visually appealing and easily understandable format	

	collection of data from the survey.		
MS Word	MS Word is a widely used word processing application that allows users to create, edit, and format text documents. Using in application report creation.	Offers extensive options for formatting text, images, and layouts, enabling the creation of professional quality documents.	
MS Power Point	MS PowerPoint is a presentation software that allows users to create dynamic slideshows comprising text, images, videos, and animations. Able us to create slideshows presentations for application.	Its intuitive interface and templates make it easy for users to create professional presentations quickly, regardless of their technical skill level	
Chrome	Google Chrome is a fast, secure, and user-friendly web browser developed by Google. Used to test our application UI	Chrome is optimized for fast page loading and efficient resource management, providing a smooth browsing experience.	
GanttProject	GanttProject is a free, open-source project management tool that	Allows users to assign tasks, set deadlines, and manage resources	



CONCEPTION AND IMPLEMENTATION OF AN INTELLIGENT SCHOOL MANAGEMENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND LEARNING ORIENTATION (CASE STUDY: ESSTIC)



	allows users to create Gantt charts for planning and scheduling projects	effectively, facilitating better project coordination and efficiency.	
--	--	---	--



CONCLUSION

In our realization phase, we set as objective to plane the different aspect of our system by showing how it will be structure and deployed within existing technical architectures. We began by taking into account the technical constraints for our system, after which we proceeded to identify the components of our system, how they are grouped together and how they should be deployed on appropriate deployment targets. We finished this phase by looking at interactions between the various aspects and actors of our system. The next phase of our report is the realization phase where we will look at aspects concerning the implementation of our system.



CHAPTER VII: FUNCTIONALITY TESTING

Preamble

This phase aims at testing the software or application to verify if the system is working as expected. It involves testing individual functionalities and capabilities of the software to ensure that they work correctly and meets the specified needs.

Content :

INTRODUCTION

I. APPLICATION MODULES

II. UNIT TESTING

CONCLUSION



INTRODUCTION

It is a systematic process that involves evaluating the functionality of a software application or system in different phases or stages of its development lifecycle. This testing approach ensures that each phase of the software development process is thoroughly tested for its intended functionalities and overall performance. Its main goal is to identify and rectify any defects, and bugs during the process.

c. APPLICTION FUNCTIONALITIES

Authentication is the fundamental functionality of the systems that verifies the identity of users(student, teacher, Admin). It ensures that only authorized individuals or entities can gain access to protected information or perform specific operations.

i. Authentication

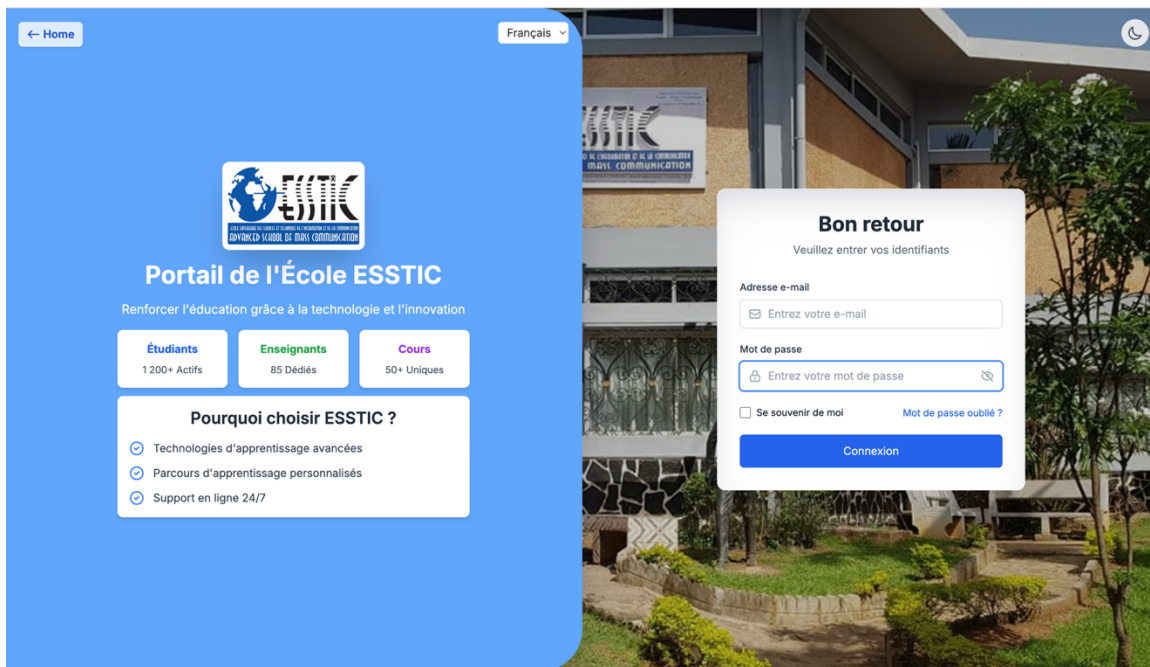


Figure 36 : Login screen

XV. RECOVER ACCOUNT

Here the user the user is able to recover his information so as to save time when registration for an exam.

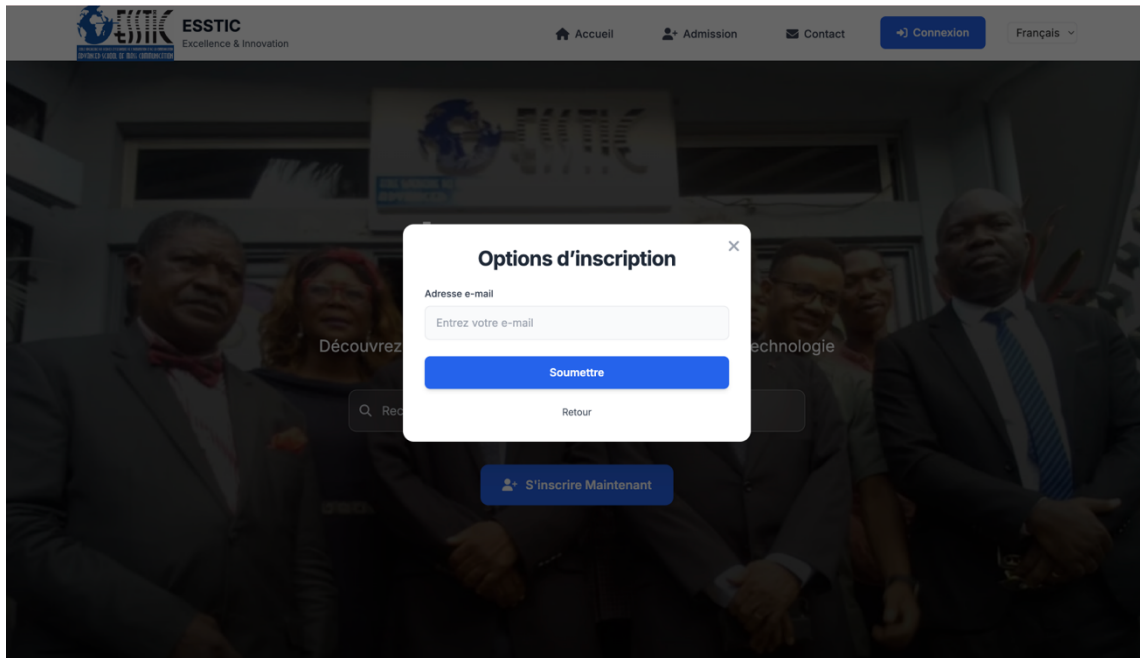


Figure 38: recover user info screen

student view his transcript

Tableau de Bord ESSTIC
Aperçu complet des performances de l'institut

Rechercher...

Transcript Preview for 2023-2024

REPUBLIQUE DU CAMEROUN
Paix - Travail - Progrès

ECOLE SUPERIEURE DES SCIENCES ET TECHNIQUES DE L'INFORMATION ET DE LA COMMUNICATION

RELEVÉ DE NOTES / TRANSCRIPT

Name: Mohamed Ousmanou nana
Registration N°: 25R0170T
Department: Bibliothèque et Doc

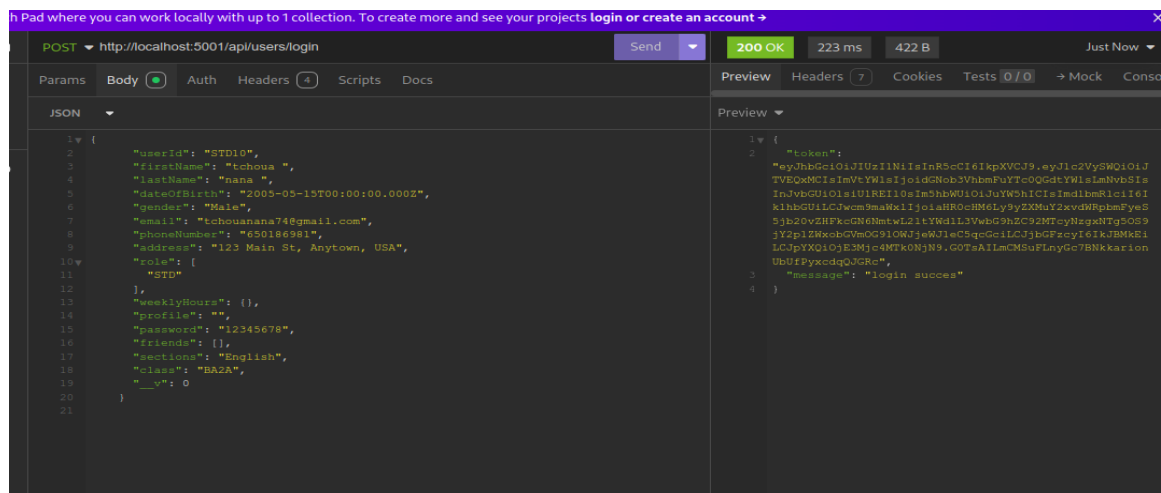
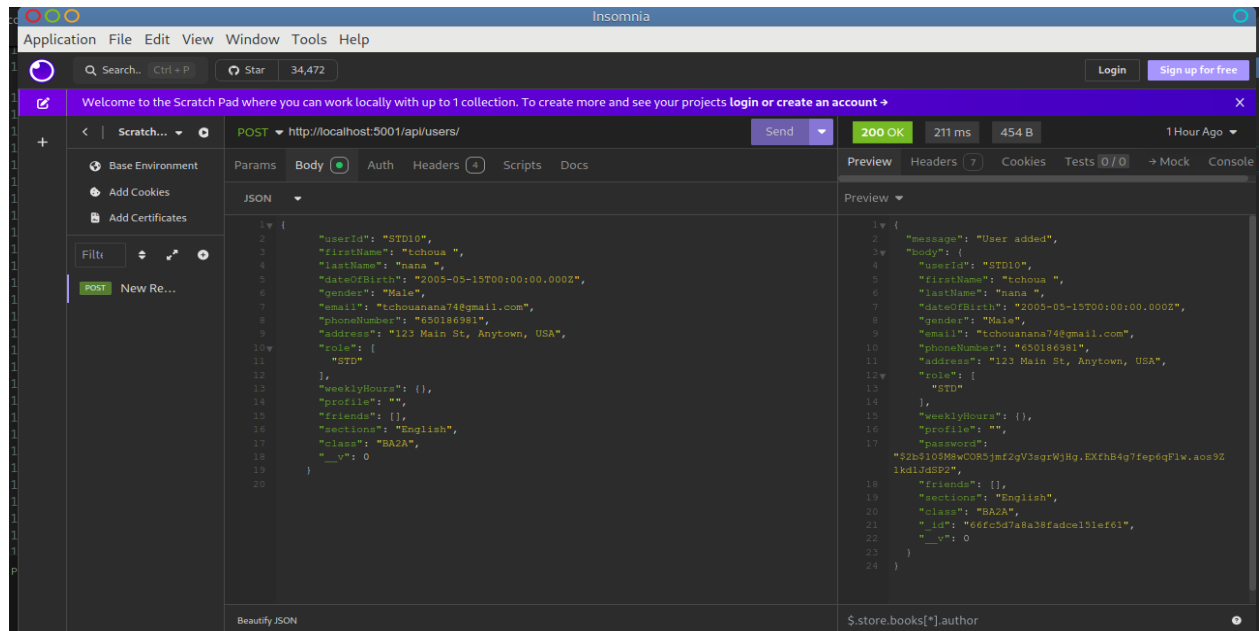
Born on: 30/05/1999
Level: Level 3
Academic Year: 2023-2024

Code	Course Title	Grade	Coef.	Mention
BDOF51-1-sem1-2024	Théorie de l'information, de la communication et du document (TRONC COMMUN)	12.60	4	Assez Bien
BDOF51-2-sem1-2024	Profession et déontologie (TRONC COMMUN)	11.80	4	Passable
BDOF52-2-sem1-2024	Gestion d'un SID 2 : RH et gestion d'équipe (TRONC COMMUN)	15.40	4	Bien
BDOF52-3-sem1-2024	Knowledge management 1 (TRONC COMMUN)	18.00	4	Très Bien
BDOF52-4-sem1-2024	Velle informationnelle (TRONC COMMUN)	16.00	4	Très Bien
BDOF52-5-sem1-2024	Analyse de la valeur et démarche qualité (TRONC COMMUN)	11.80	4	Passable
BDOF52-6-sem1-2024	Conservation du patrimoine documentaire (TRONC COMMUN)	8.00	4	Echec
BDOF53-3-sem1-2024	Création des bases de données documentaires	14.80	4	Bien
BDOF53-4-sem1-	Module documentaire Bibliothèque FIDUC COMBIA	14.80	4	Bien

Page-1



Unit testing refers to testing a function individually, to ensure that it is working properly. Here are some unit test of our solution:





CONCLUSION

Testing is an integral part of the software development life cycle and is performed to verify the correctness, reliability, and robustness of the implemented functionality. the functionality and testing phase are therefore a critical stage in software development where the implemented features are validated, and any issues are identified and resolved.



**CONCEPTION AND IMPLEMENTATION OF AN INTELLIGENT SCHOOL
MANAGEMENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND
LEARNING ORIENTATION (CASE STUDY: ESSTIC)**





CHAPTER VII: INSTALLATION GUIDE AND USER GUIDE

Preamble

A user guide explains how to use a software application in a language that a non-technical person can understand. Thus, it enables the user to easily use the application to familiarize with the software and discover all its functionalities.

INTRODUCTION

I. DESCRIPTION OF THE APPLICATION

II. INSTALLATION GUIDE

III. USER GUIDE

CONCLUSION



INTRODUCTION

The user manual is the document established after the realization of an application, software or platform. It informs on the question "how to use the software, the application or the platform that you have in front of you? ". It is therefore crucial for us to offer the various users of our platform a guide allowing them to easily perform various operations on our platform.

I. DESCRIPTION OF THE APPLICATION

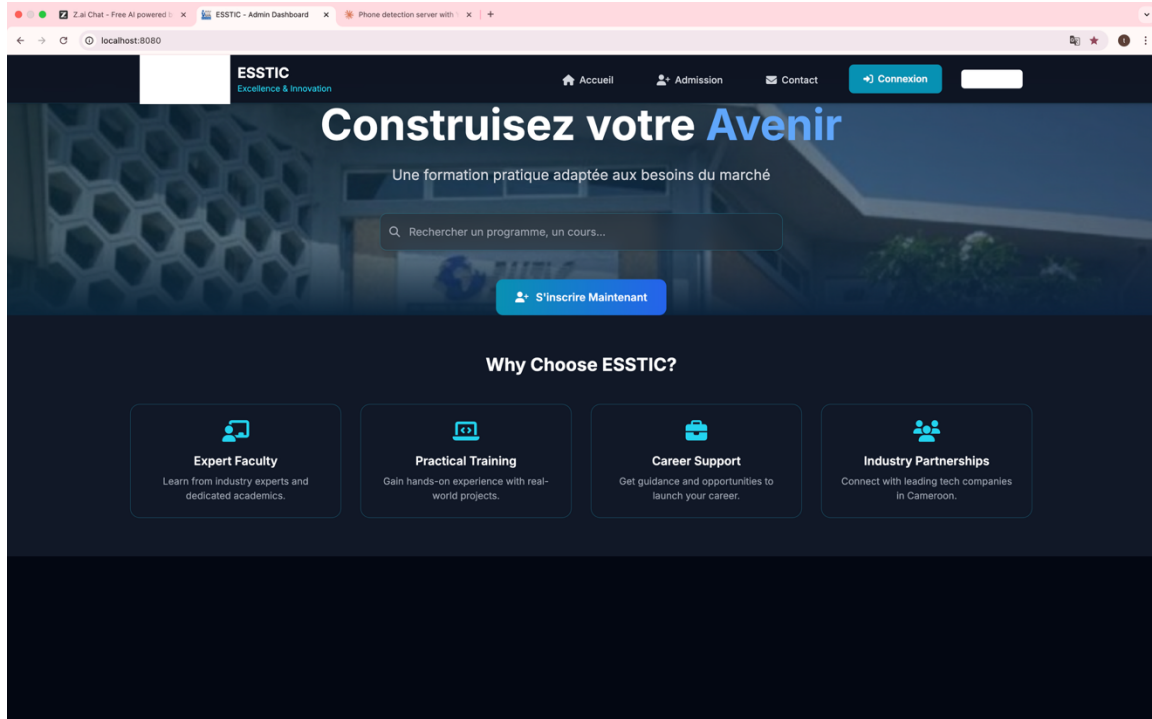
ESSTIC Smart System is an intelligent school management platform built with **Vue.js** for an interactive and responsive user interface and **Node.js** for a robust, scalable backend. The system is designed to streamline communication, scheduling, and academic performance analysis within the ESSTIC community.

This cross-platform solution centralizes academic data, enabling students, teachers, administrators, and parents to interact in real time. It provides:

- II. **Automated scheduling** for classes, exams, and events.
- III. **Intelligent performance analysis** using data-driven insights to track student progress.
- IV. **Learning orientation tools**, offering personalized recommendations and guidance.
- V. **Secure messaging and notification system** to foster transparent communication.

INSTALLATION GUIDE

To be able to ESSTIC web application, we just need a stable internet connection and go to a navigator, then type the following URL 'https://esstic.cm' then the user will be redirected to the landing page as follow:



VI. USER GUIDE

In this part, the steps of carrying out the operations will be explained in detail. All this will be done through tests and explanations with screenshots in support to allow users not only to know how to actually do manipulations, but also to allow them to have an overview of the results they should expect.

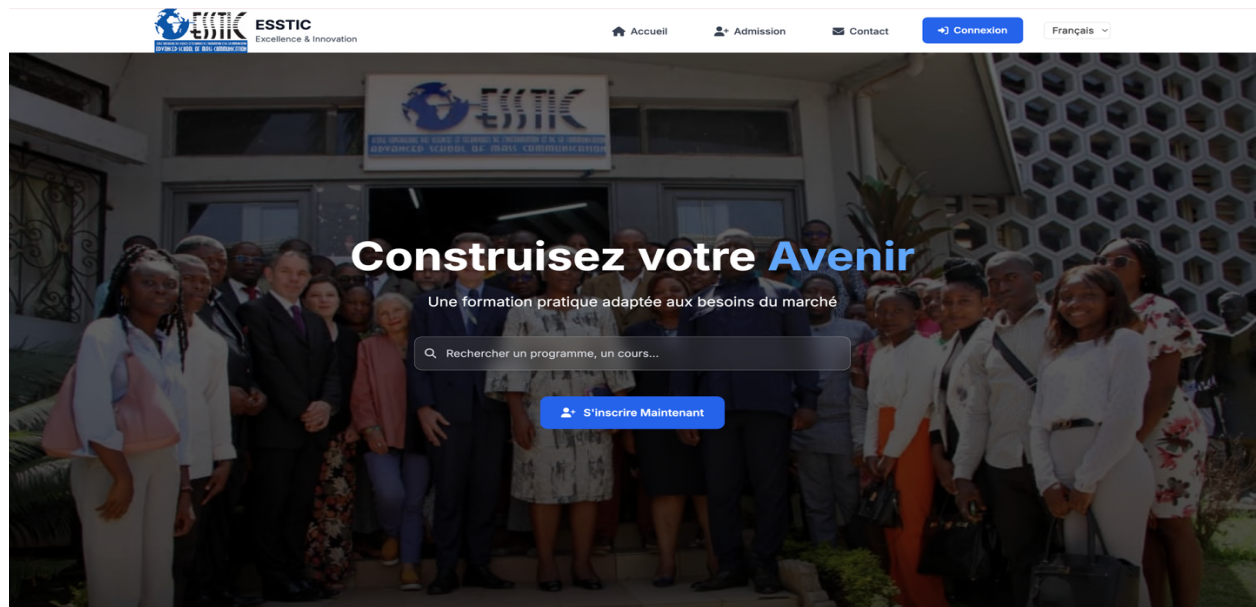
1. Connecting to the platform

Connecting to the platform is not complicated at all. However, it is essential to meet a number of prerequisites, namely:

- ✓ Have a computer, tablet or smartphone;
- ✓ Have installed a web browser;
- ✓ Have Access to the Internet

You will then obtain this page

1. Landing Page:



2.Login Page

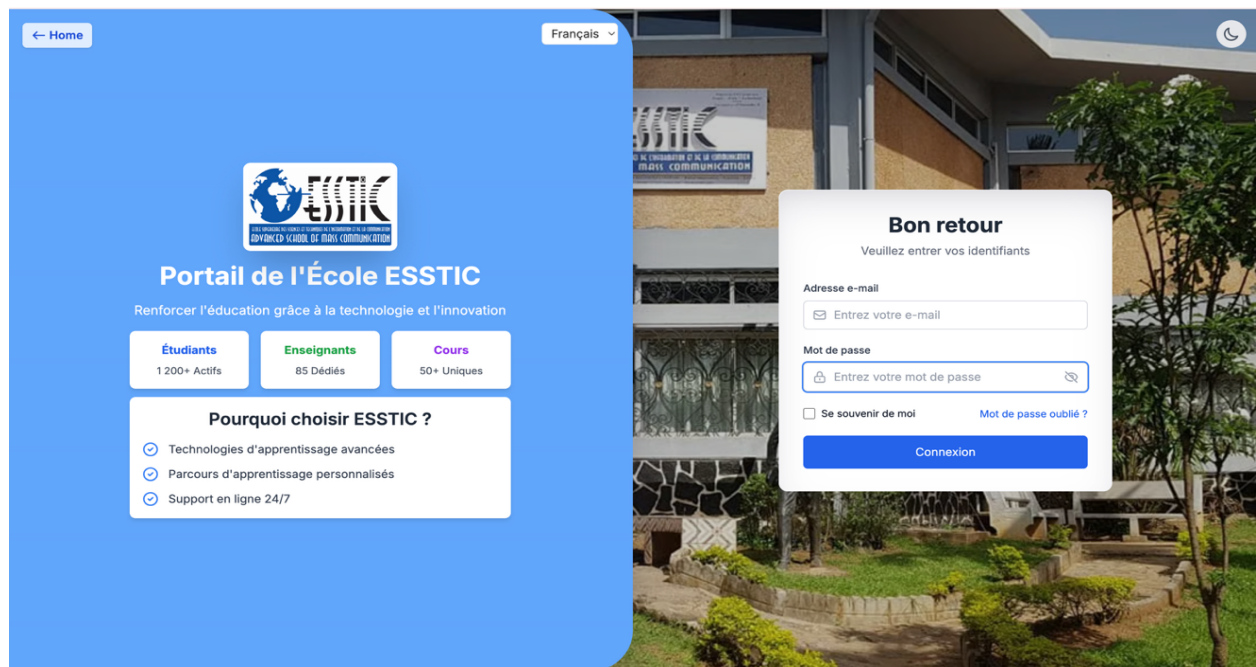


Figure 41:login page



CONCEPTION AND IMPLEMENTATION OF AN INTELLIGENT SCHOOL MANAGEMENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND LEARNING ORIENTATION (CASE STUDY: ESSTIC)



Dashboard:

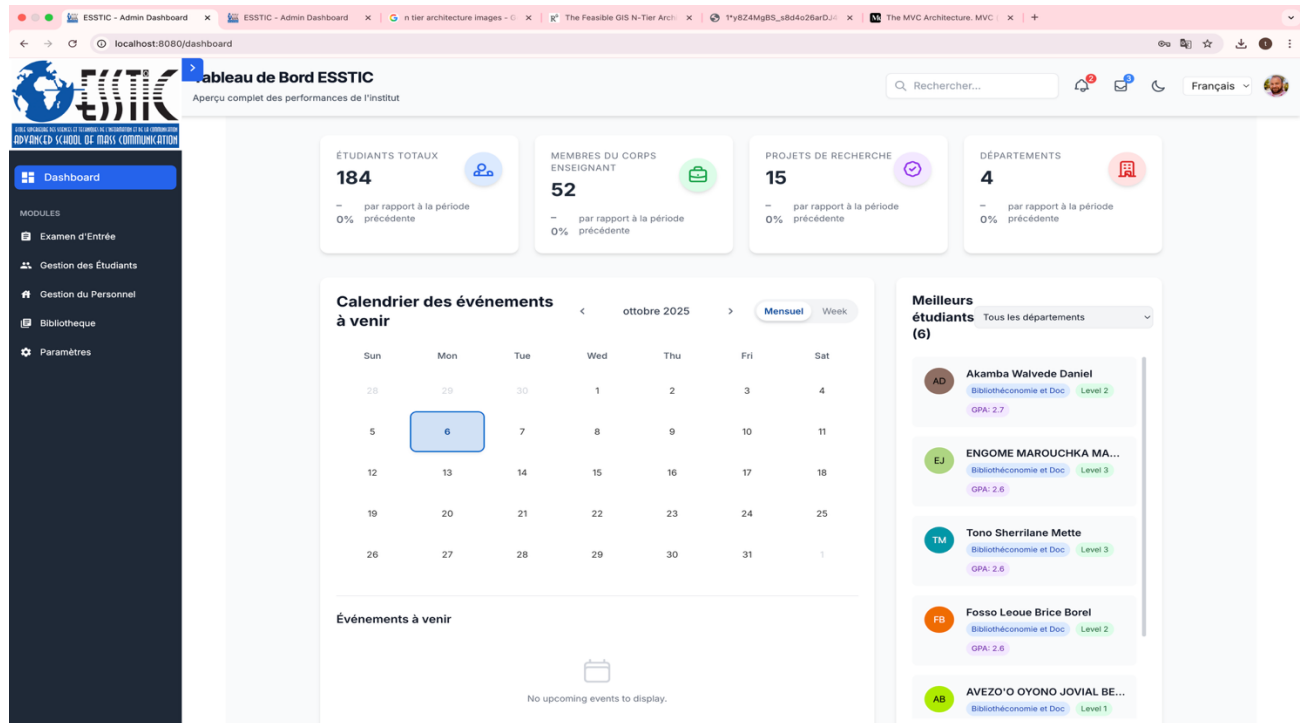


Figure 42: dashbord interface

Validate candidate interface:

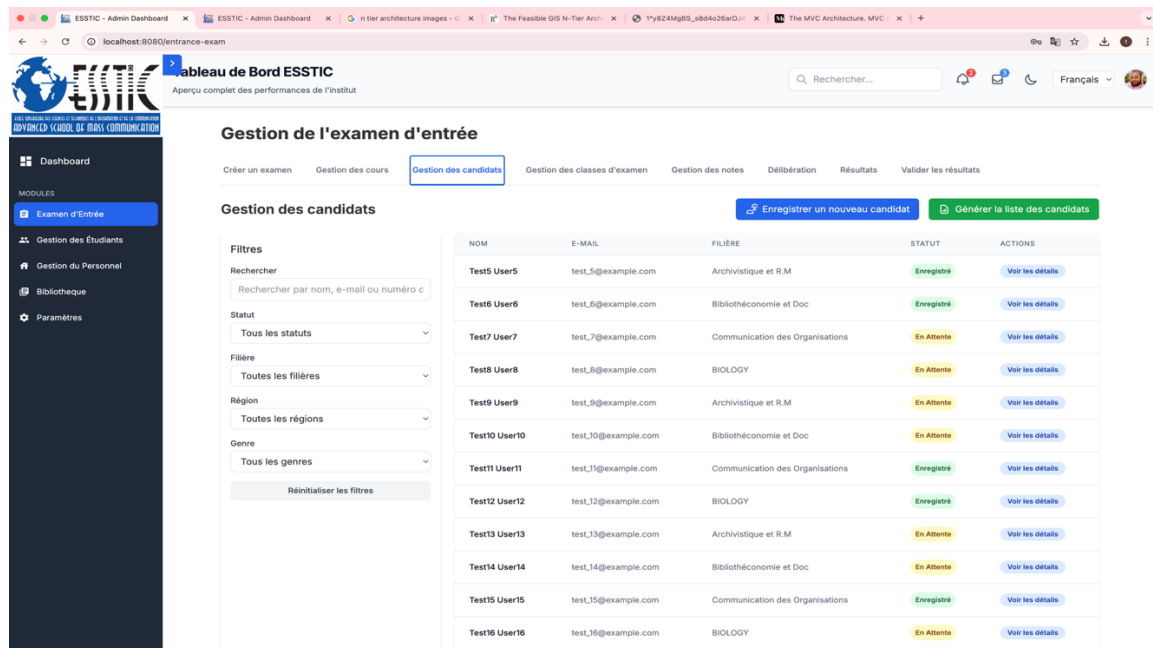


Figure 43: view timetable

Validate entrance exam result

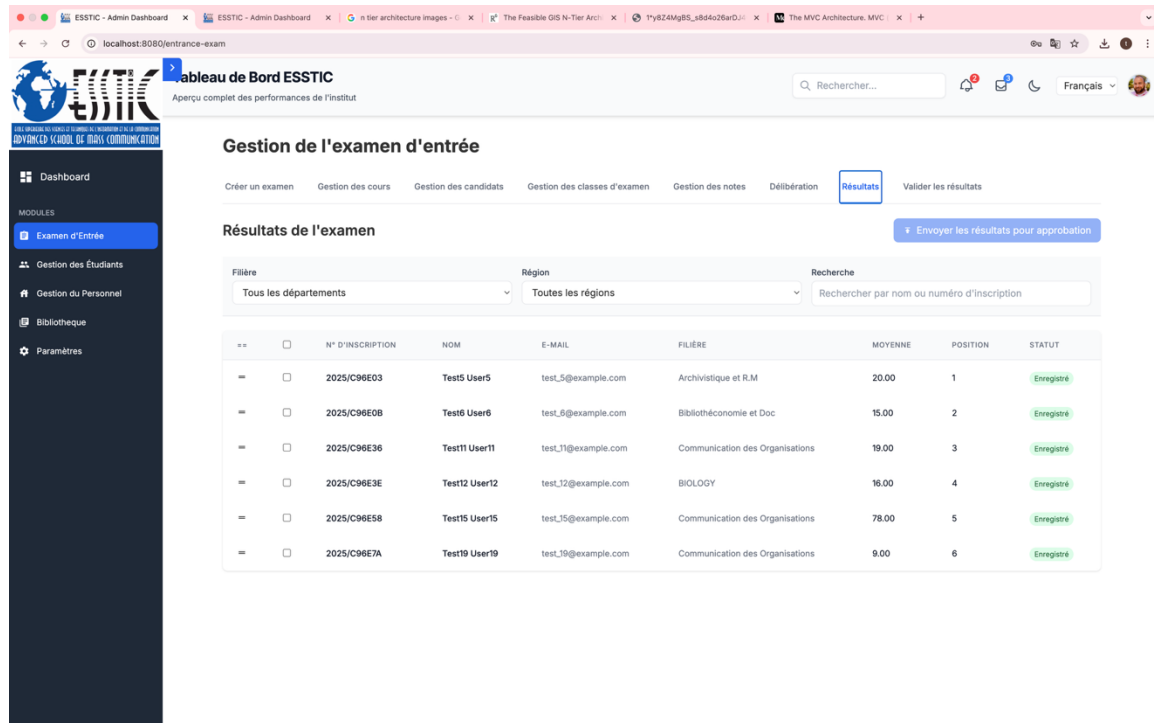


Figure 44:validate entrance exam result

Manage Books

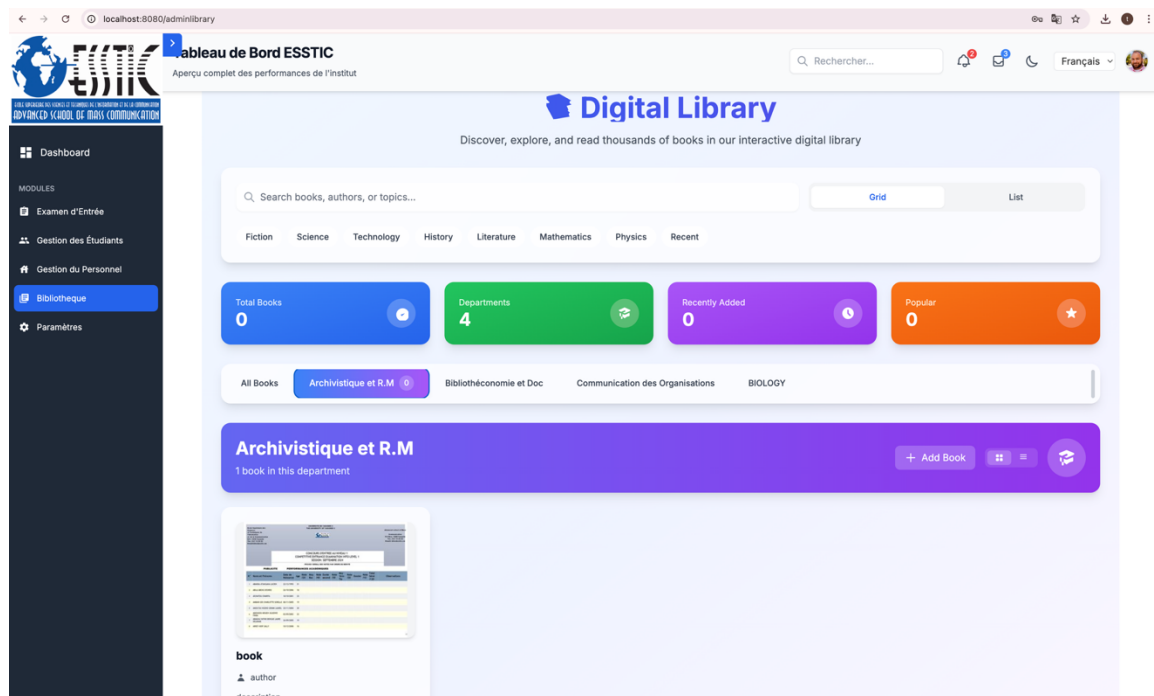


Figure 45:manage profile interface



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Manage permissions:

Settings Management

Permission Management

9 permission(s)

Permission	Status
read	Active
export	Active
import	Active
delete	Active
update	Active
generate	Active
request	Active
create	Active
publish	Active

Manage Roles:

Settings Management

Role Management

+ Add Role

Name	Status	Actions
Admin	Active	Edit Deactivate Delete
Teacher	Active	Edit Deactivate Delete

GENERAL CONCLUSION

As we bring this project to a close, we take pride in the knowledge acquired, the challenges overcome, and the milestones achieved throughout its development. The journey of conceiving, analyzing, and designing the **Intelligent School Management System for Student Performance Analysis and Learning Orientation** has been both intellectually stimulating and professionally rewarding. It has allowed us to apply theoretical concepts to practical scenarios, strengthening our understanding of software engineering principles and project management practices. Guided by the **2TUP (Two-Track Unified Process)** methodology, we structured our work through clear phases of analysis and conception, ensuring coherence between user requirements and system functionalities. The use of **UML (Unified Modeling Language)** provided a rigorous framework for modeling system behavior, user interactions, and data flow, thereby reinforcing the analytical precision of our design. The implemented system, developed using **Vue.js** for the front-end and **Node.js** for the back-end, integrates **artificial intelligence techniques** to analyze student performance and deliver personalized learning recommendations. It not only enhances administrative efficiency but also promotes informed decision-making, continuous academic monitoring, and personalized educational support for students. In conclusion, this project represents a meaningful contribution to the digital transformation of education. It demonstrates how modern technologies and structured methodologies can be combined to create intelligent, user-centered solutions that improve learning outcomes and institutional management. We are confident that, with further refinement and deployment, this system will serve as a valuable tool for educational institutions seeking innovation and excellence in academic performance management.



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ANNEXES



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